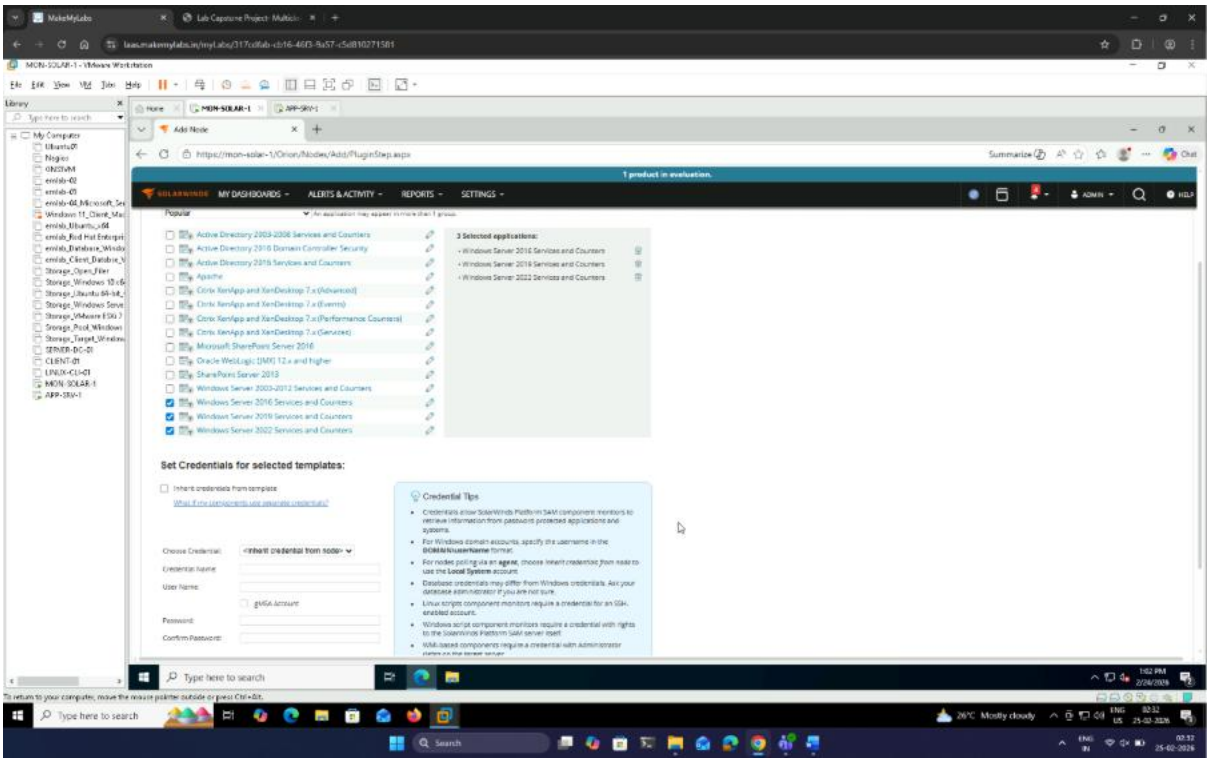
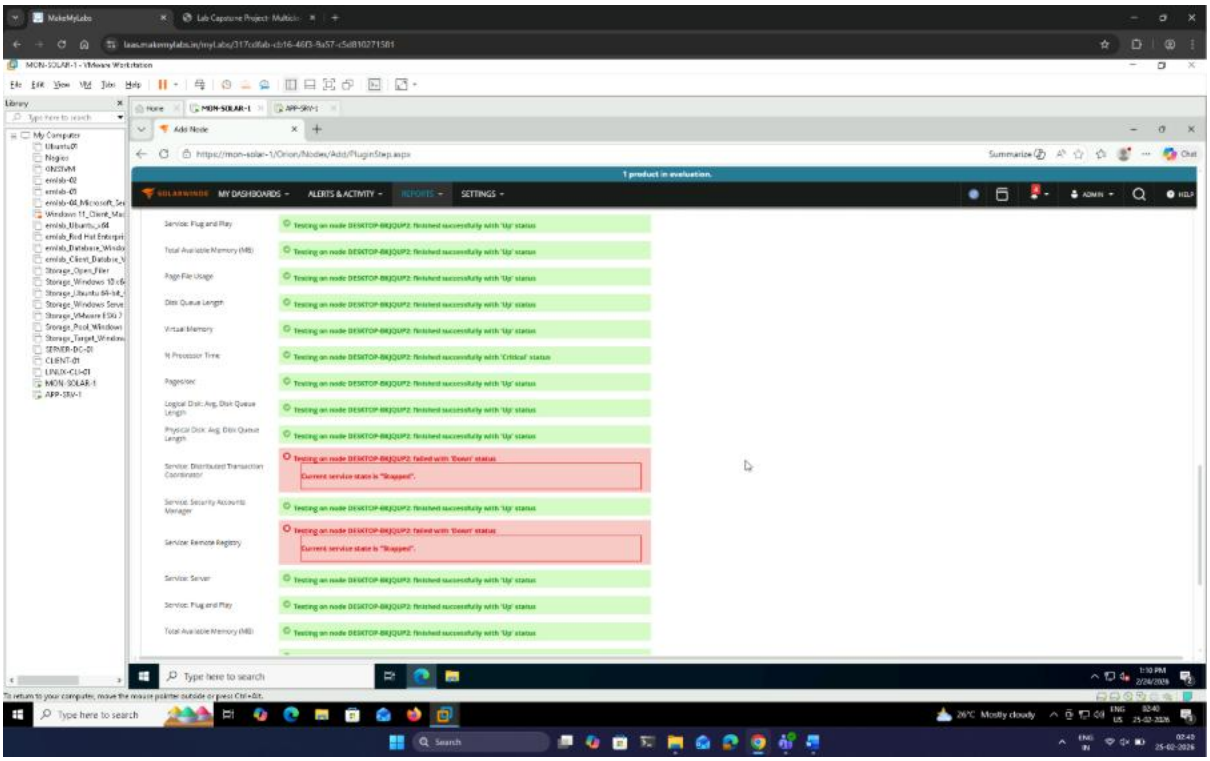
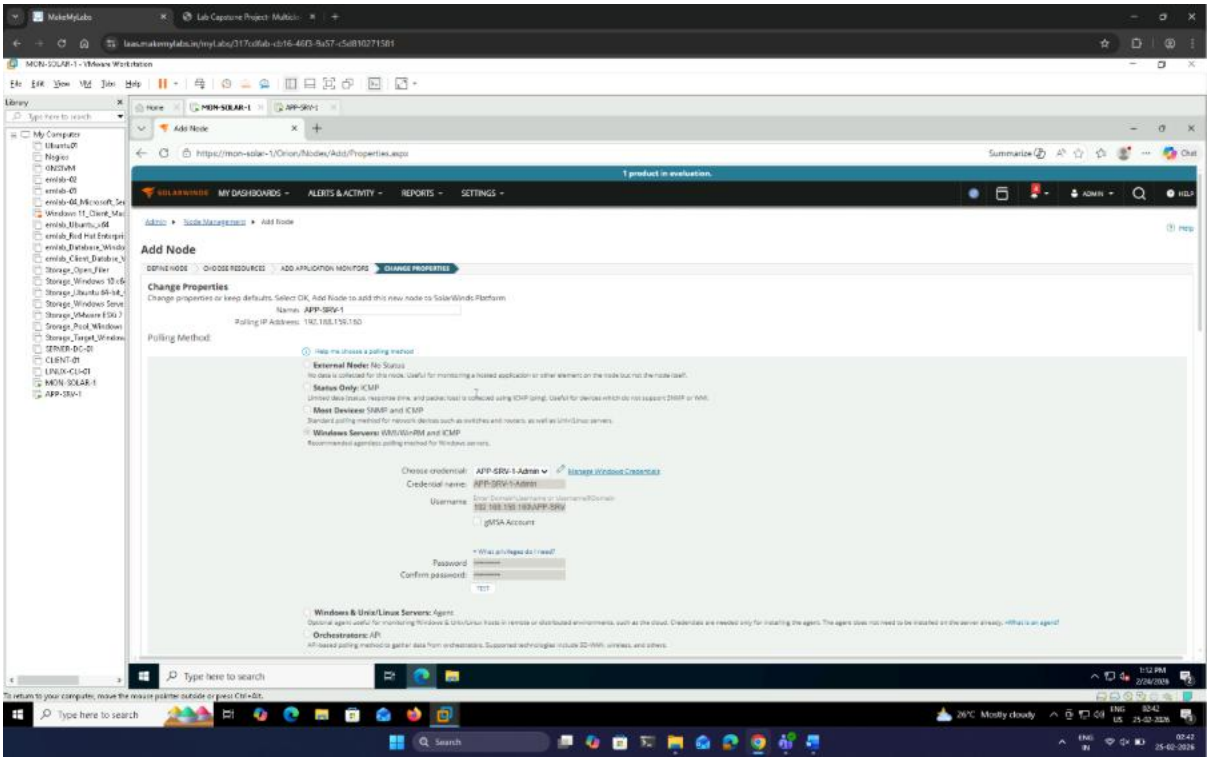
After the node was successfully added, SolarWinds automatically discovered system resources available for monitoring. Key components such as CPU Utilization, Memory Usage, Disk Volumes, and network interfaces were selected to enable infrastructure-level performance monitoring. This configuration allows real-time tracking of system health and resource consumption.

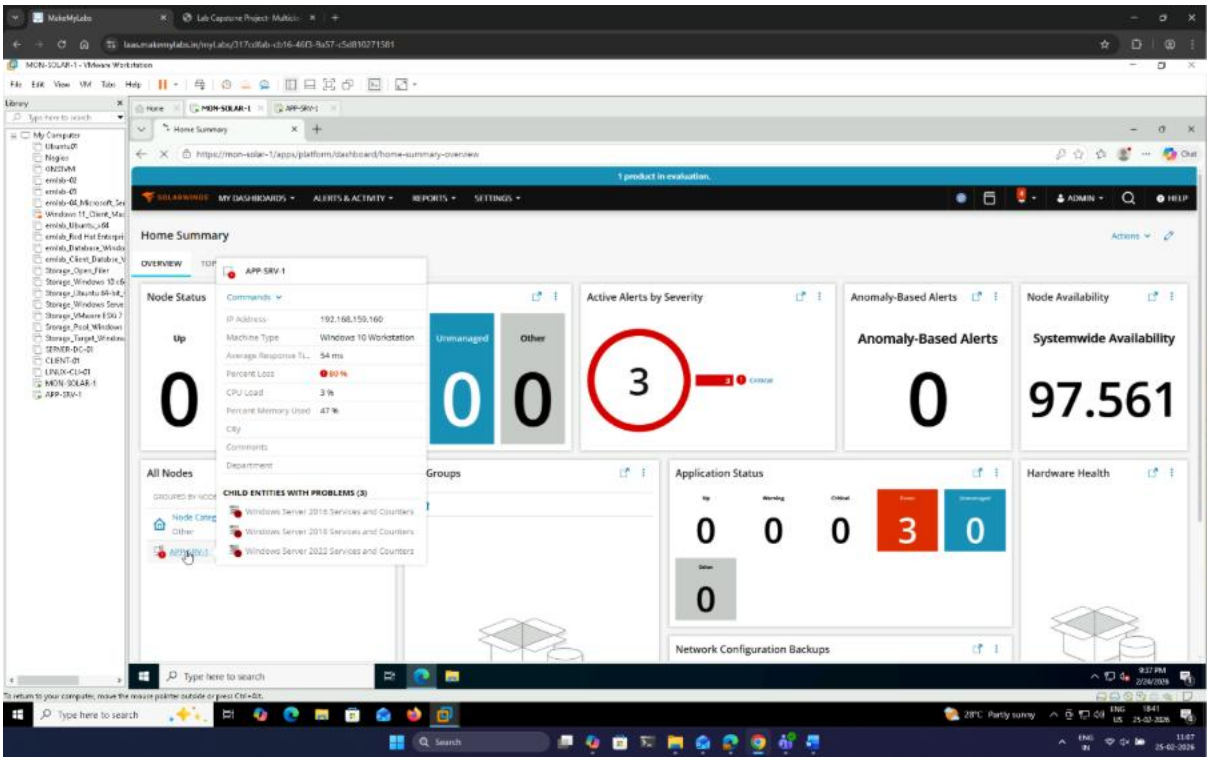


Predefined SolarWinds application templates were reviewed.

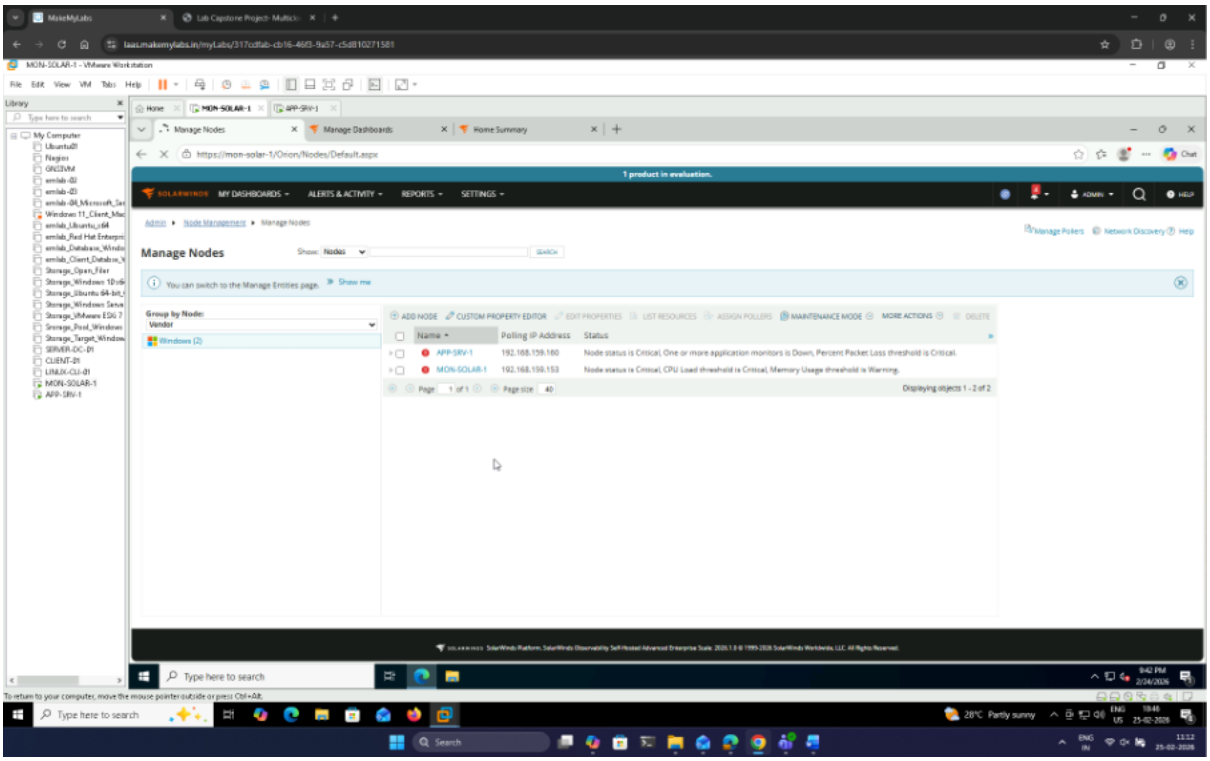


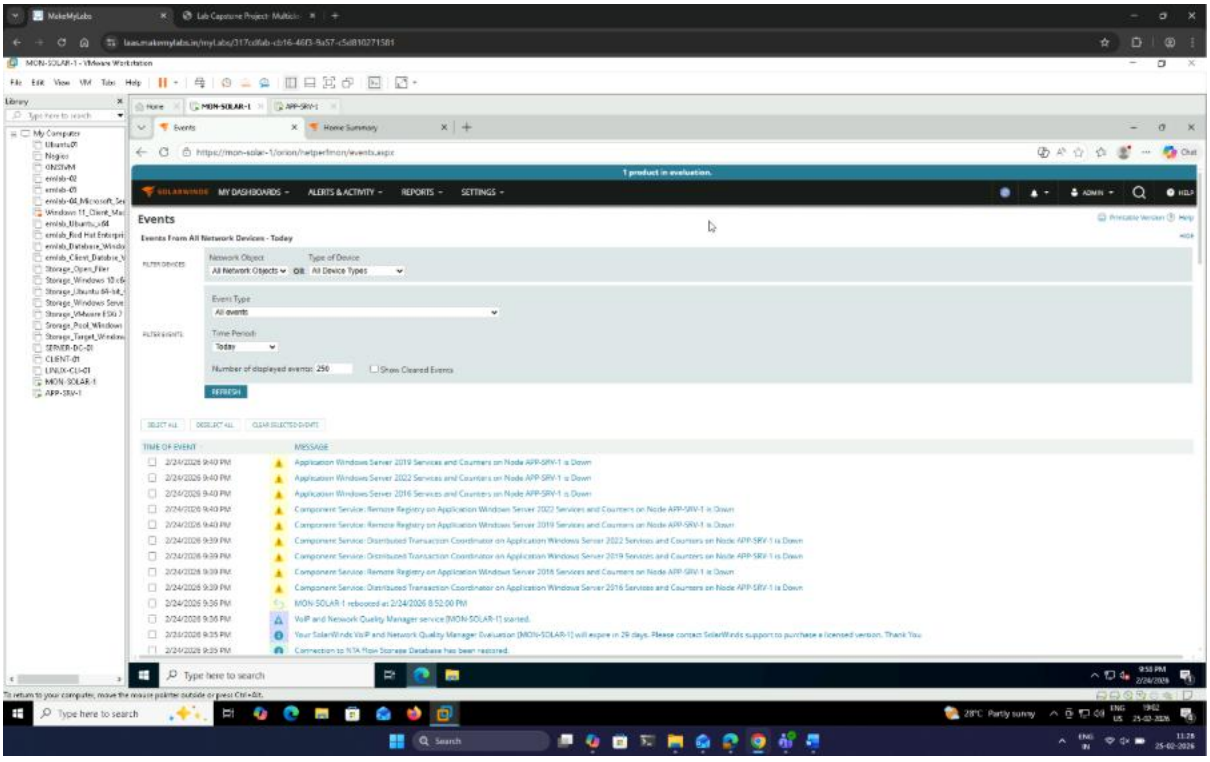
All selected components were tested first.



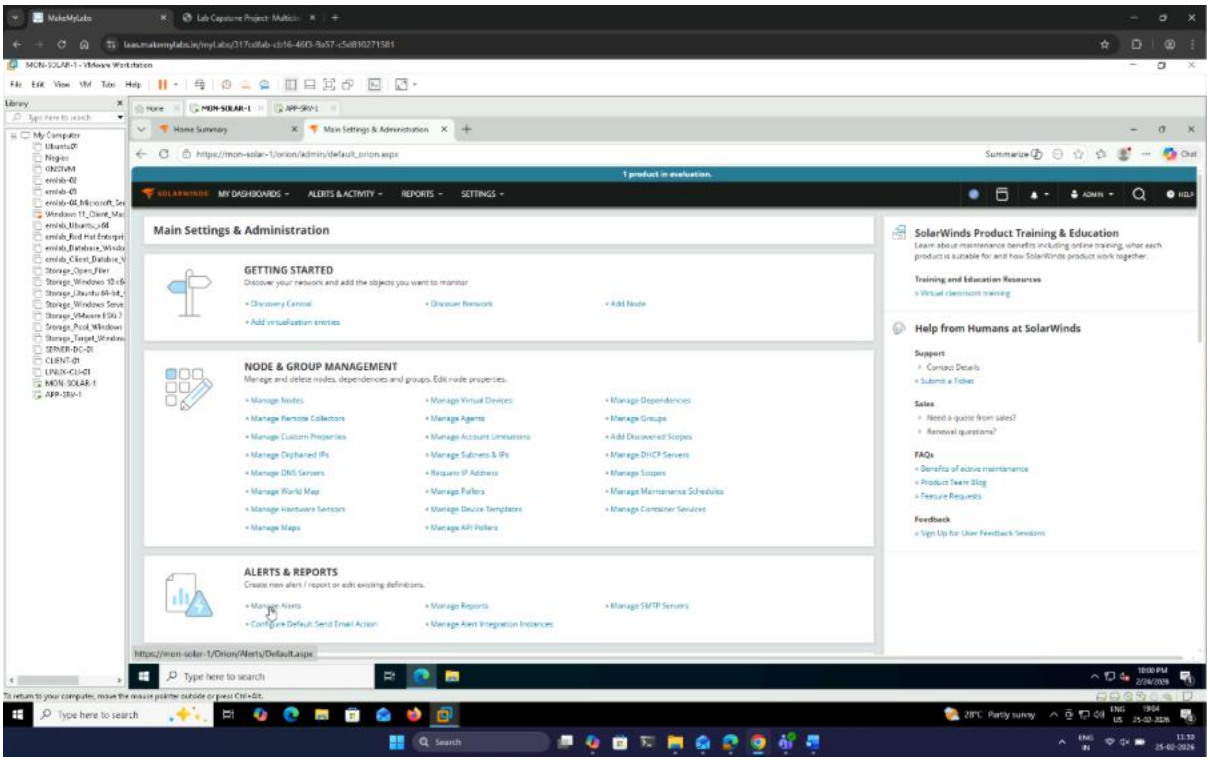


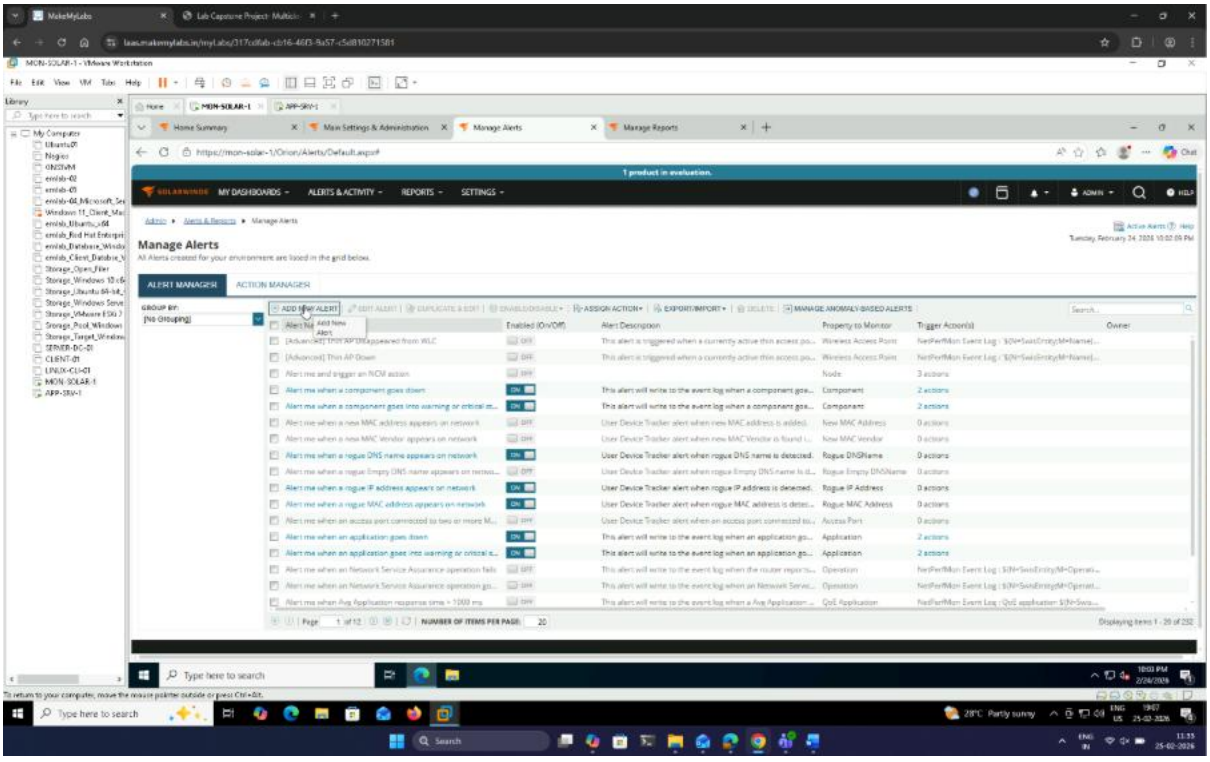
After configuration, the monitored node was verified through the SolarWinds dashboard. Performance for CPU, Memory. Disk utilization and network interfaces were successfully displayed. Application monitoring components showed real-time service status, confirimg that agentless monitoring via WMI/WinRM was functioning correctly.



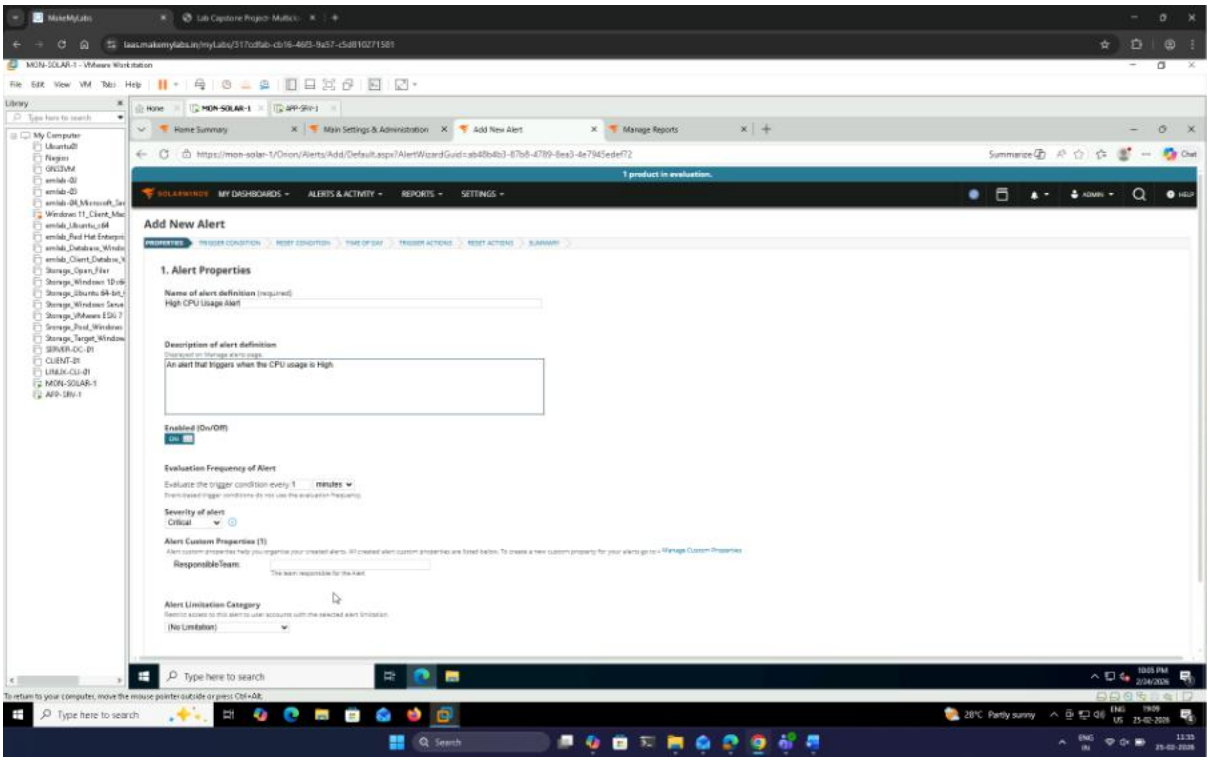


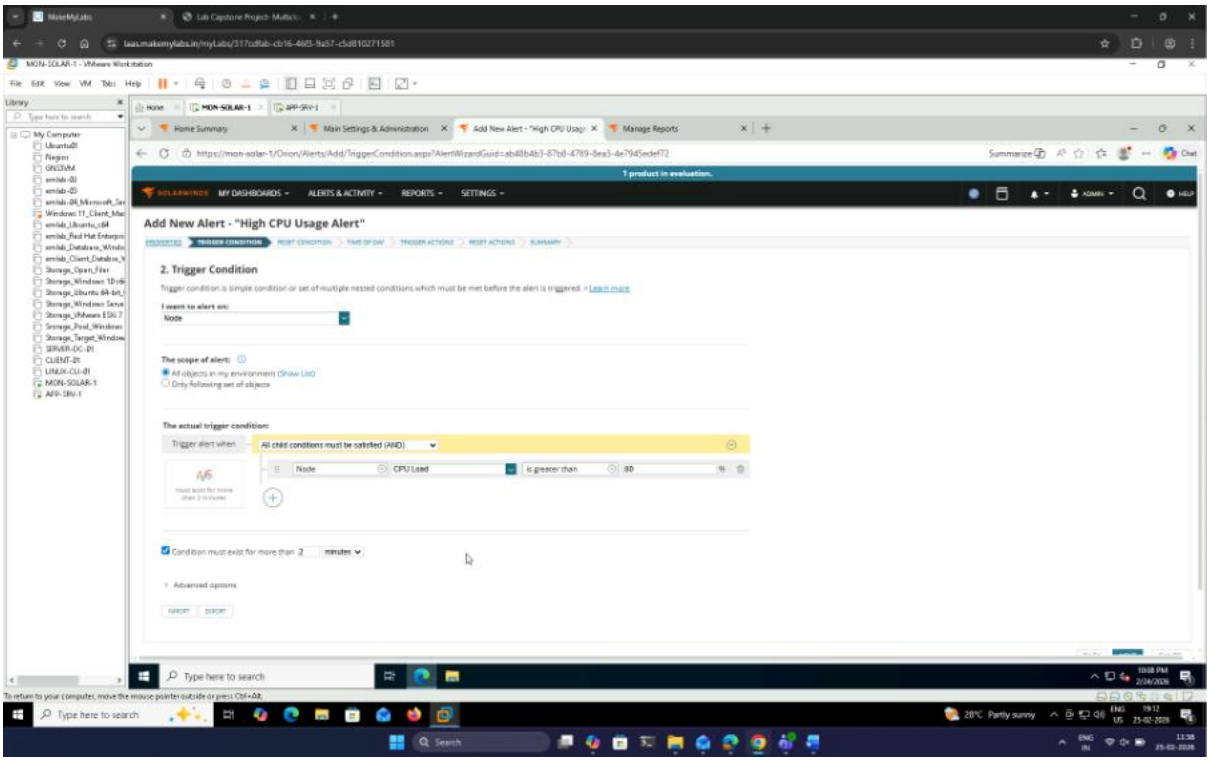
All the events have been recorded.



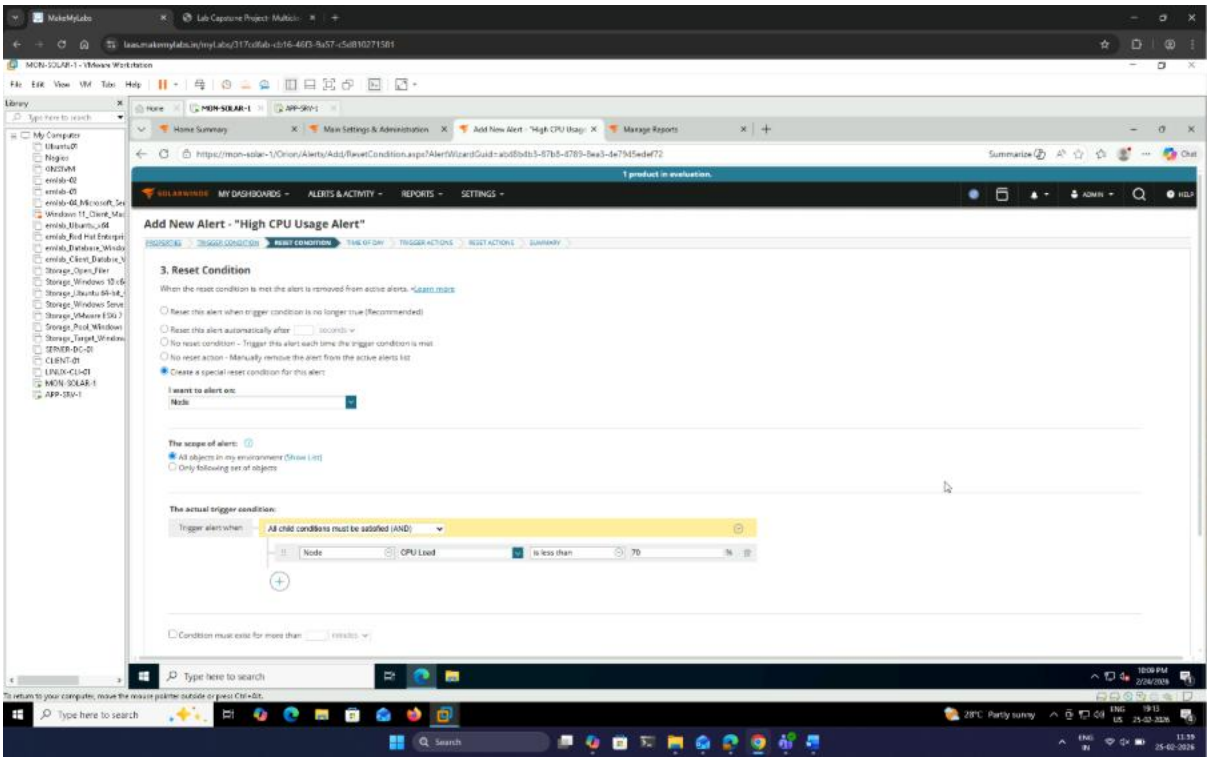


To configure proactive monitoring, the Alert Manager section of SolarWinds was accessed through the All Settings menu. A new alert was created to monitor system performance thresholds and generate notifications when pre-defined conditions were met.

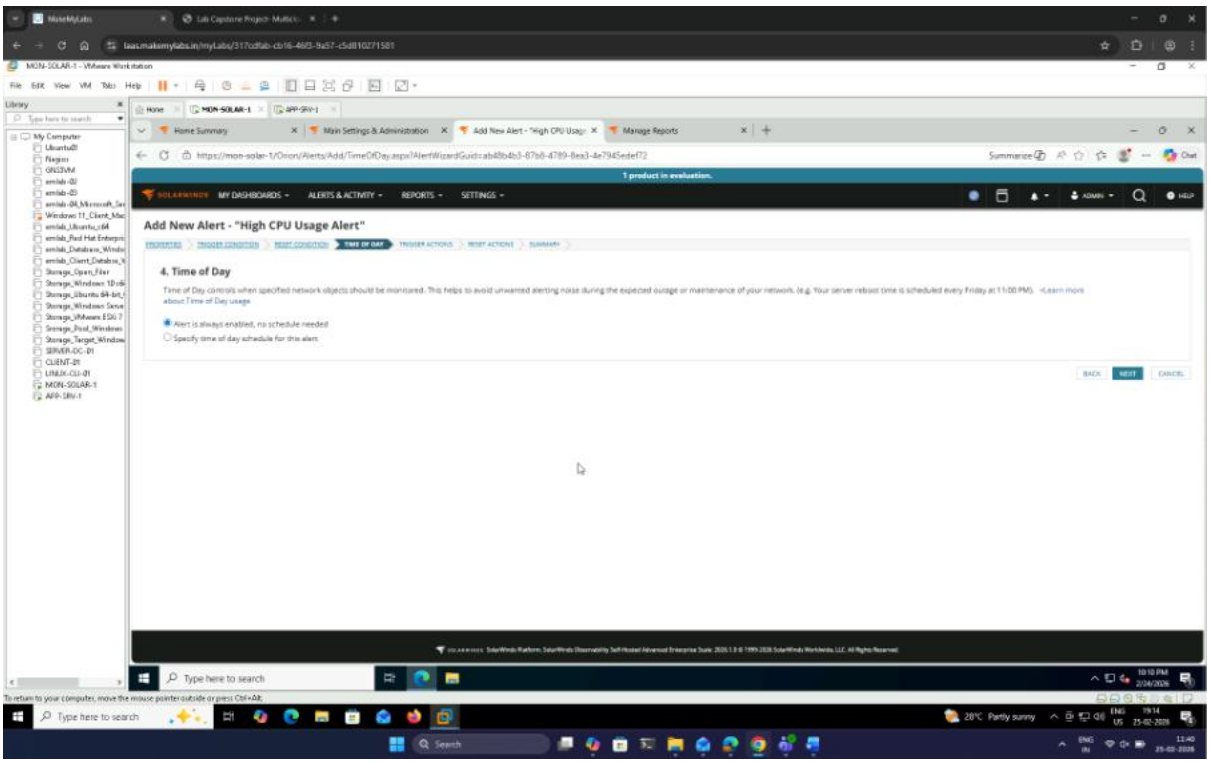


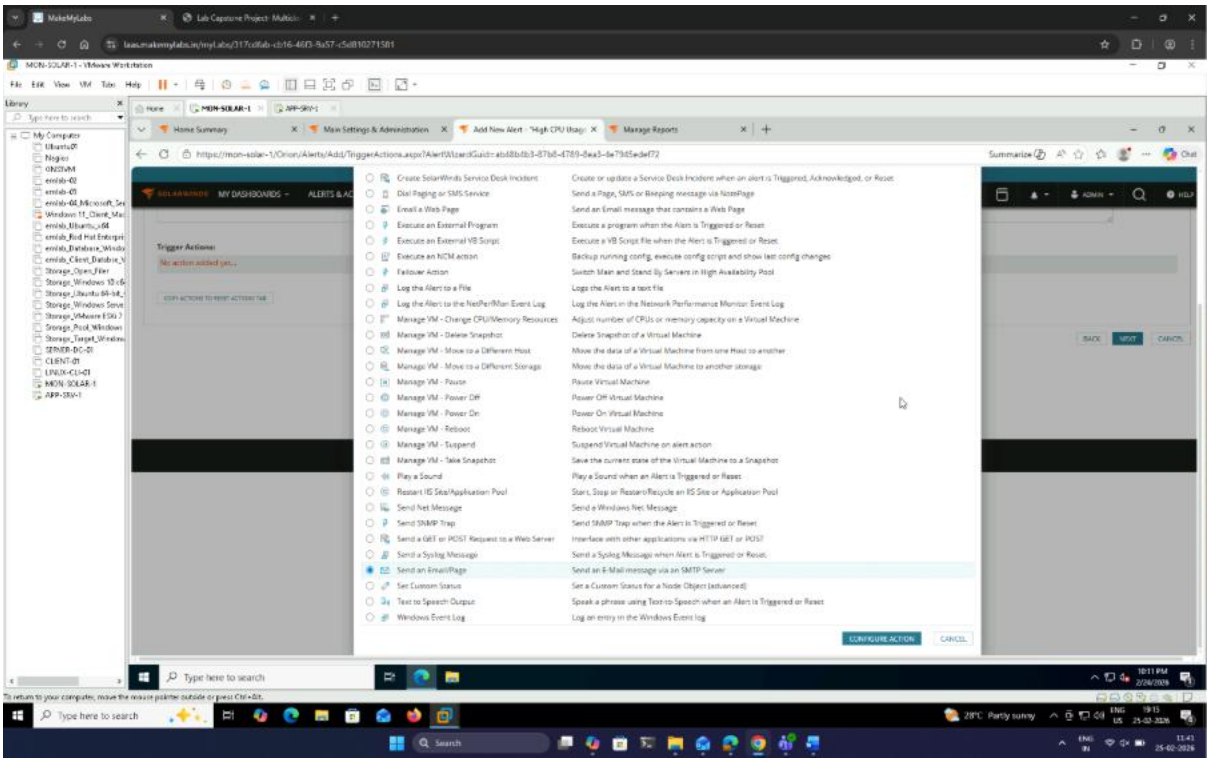


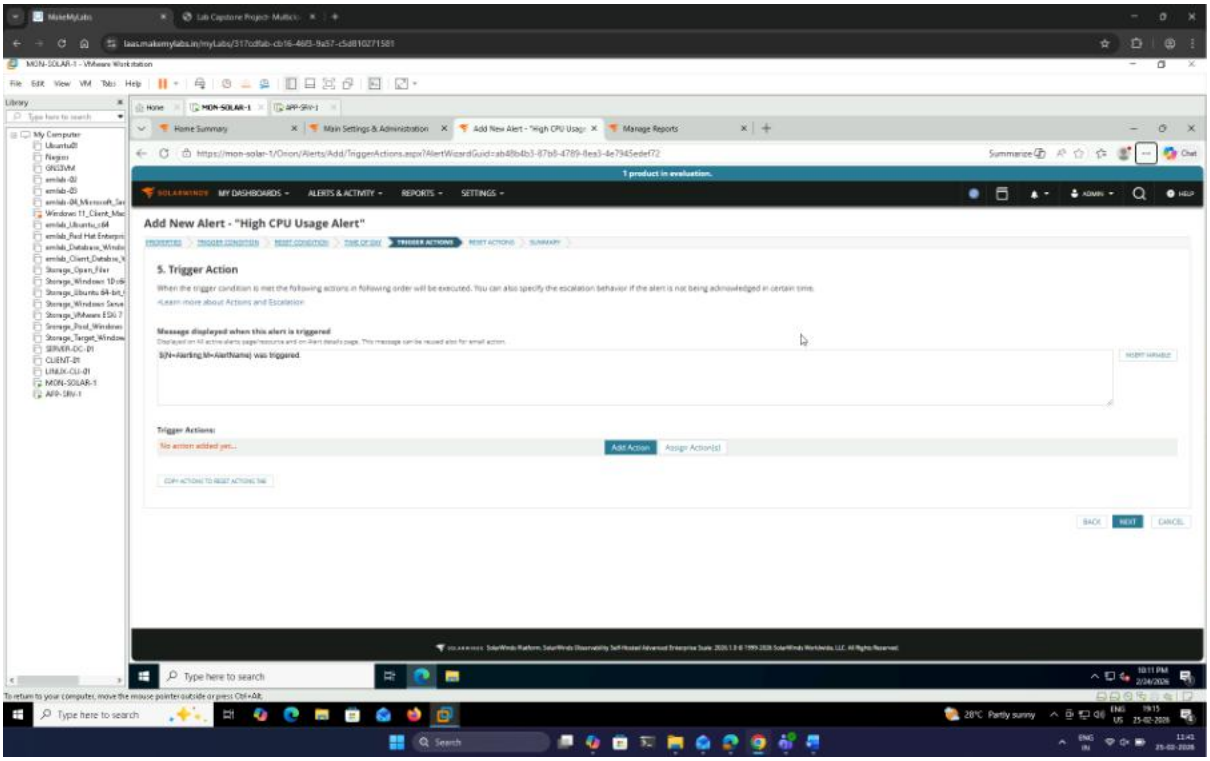
A trigger condition was configured to activate the alert when the node CPU load exceeds 80% for more than 2 minutes. This ensures that temporary performance spikes do not generate false alarms.



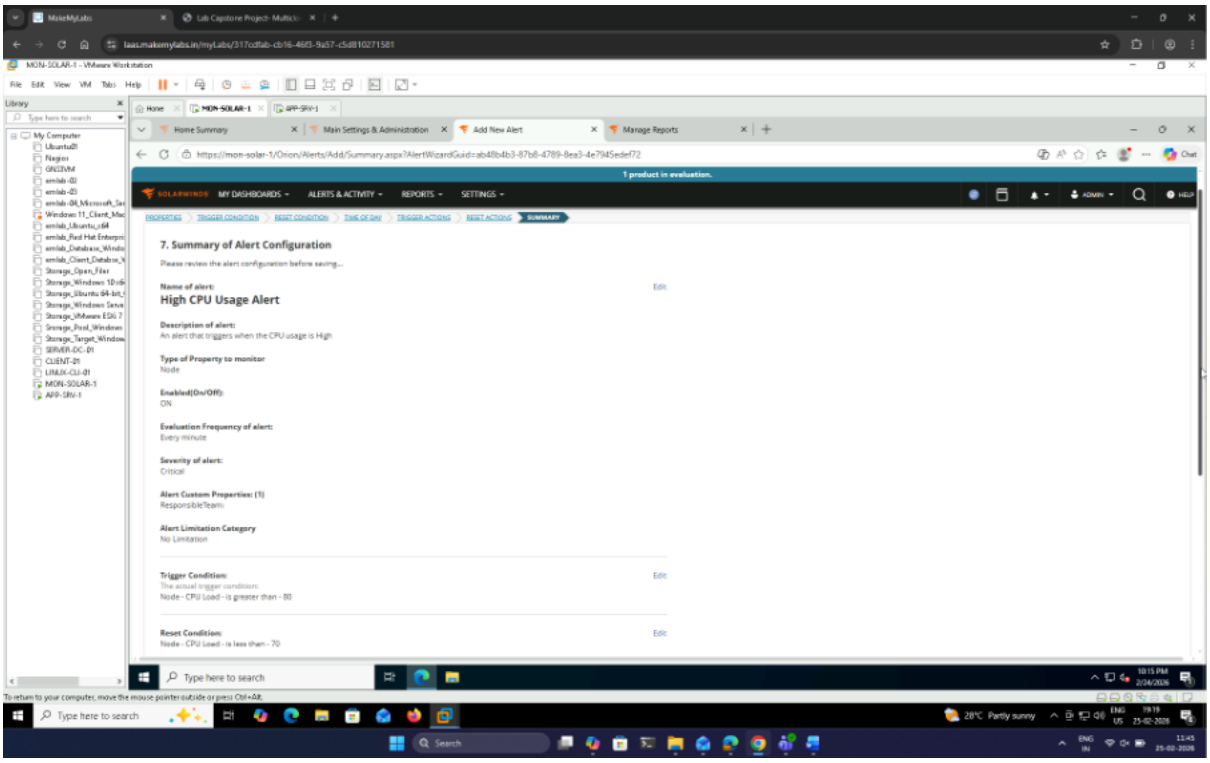
A reset condition was configured to automatically clear the alert once CPU usage falls below 70%. This allows SolarWinds to close the alert when system performance returns to normal levels.

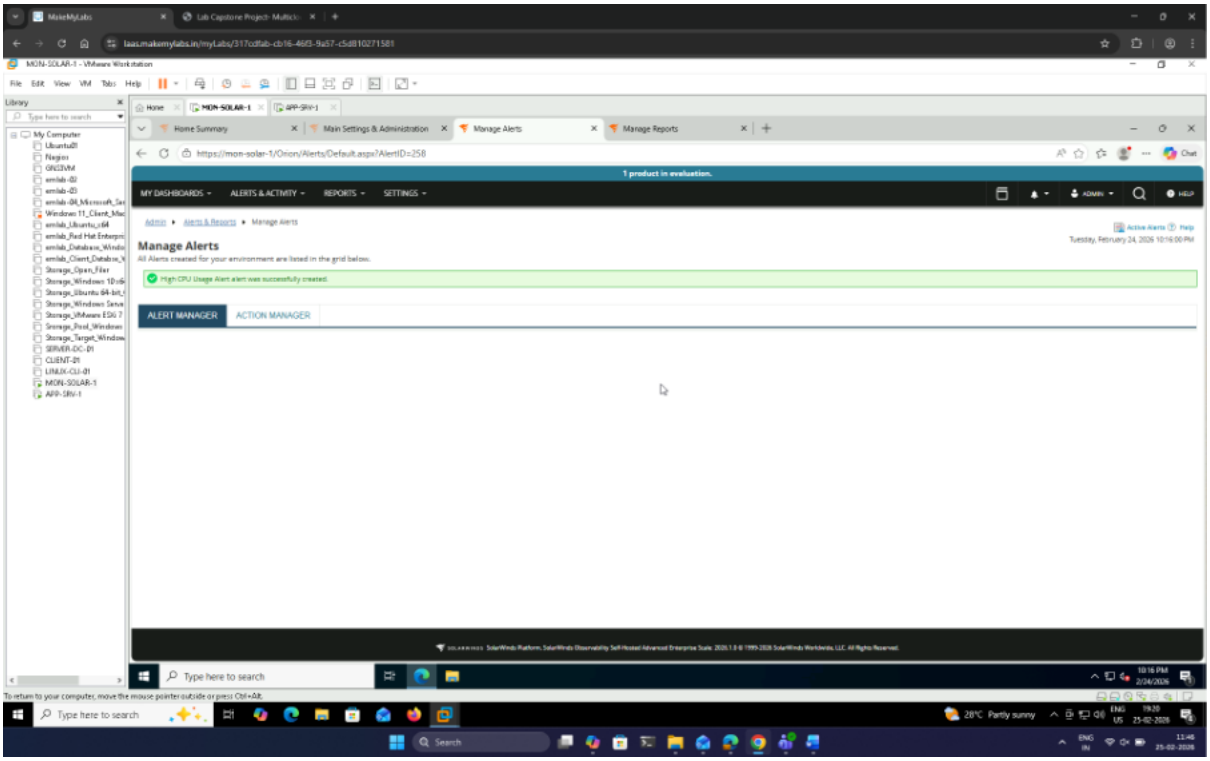


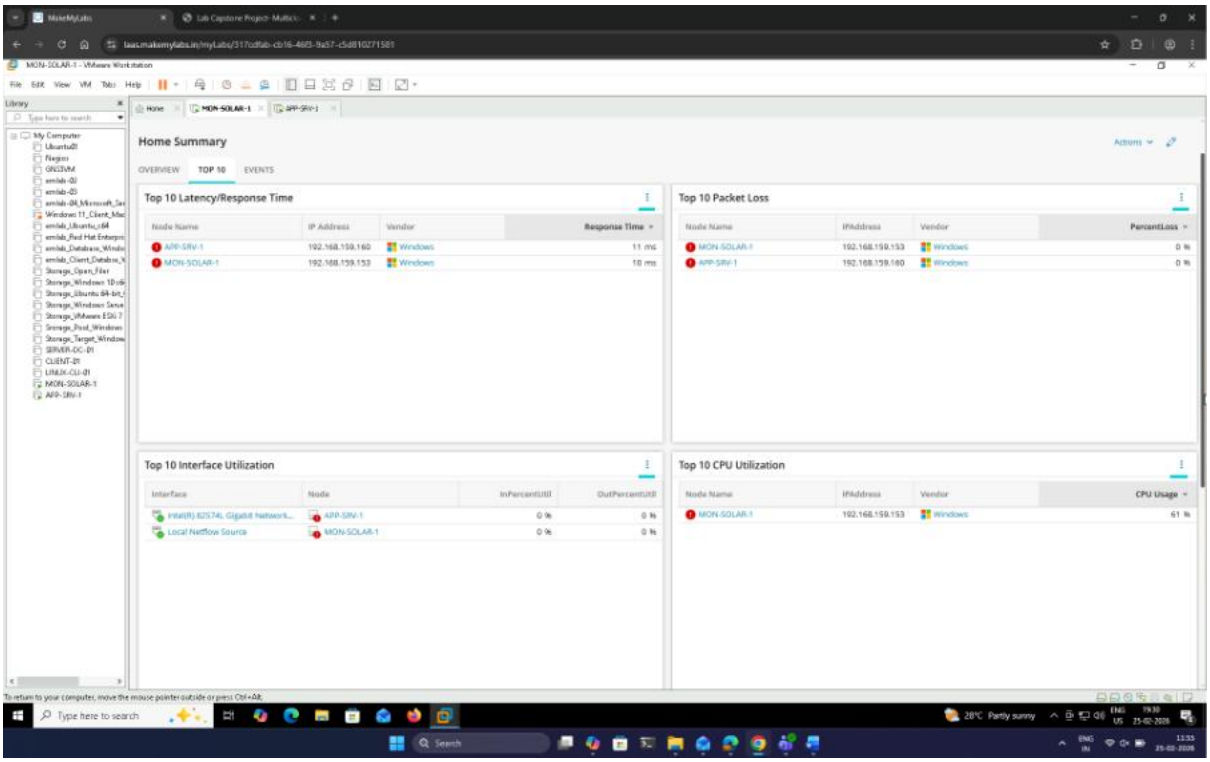

Add New Alert - "High CPU Usage"
Trigger Actions:
No action added yet...
CONFIGURE ACTION
CANCEL

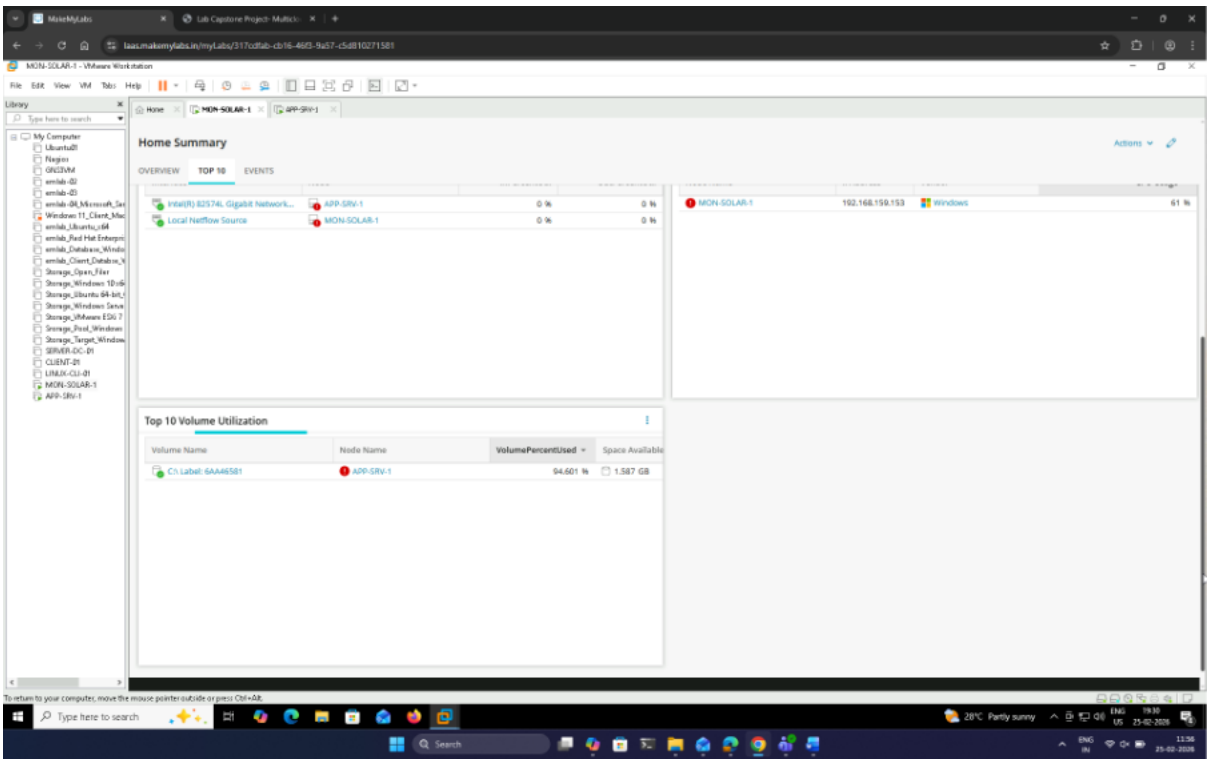

Add New Alert - "High CPU Usage Alert"
5. Trigger Action
Trigger Actions:
No action added yet...
Add Action
BACK
NEXT
CANCEL

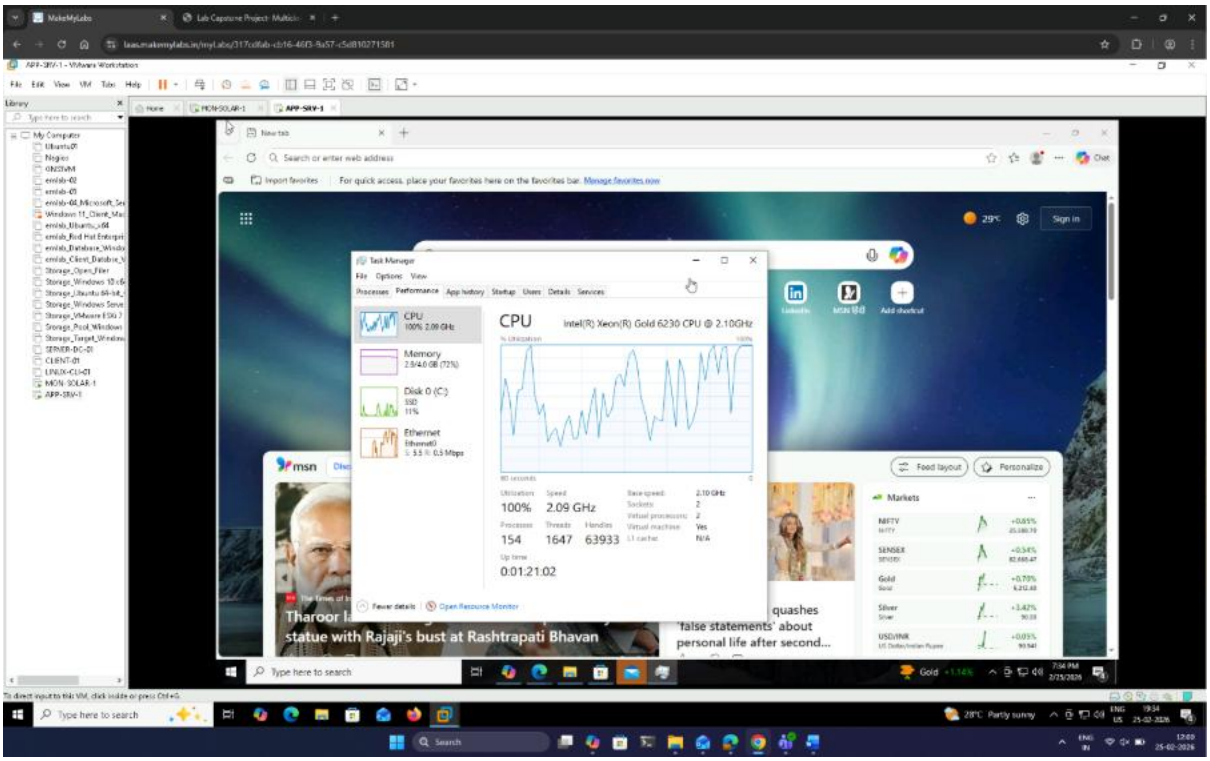
An email notification action was configured to inform administrators when the alert condition is triggered. The notification includes dynamic variables such as node name to provide contextual information. This enables proactive response to performance issues.

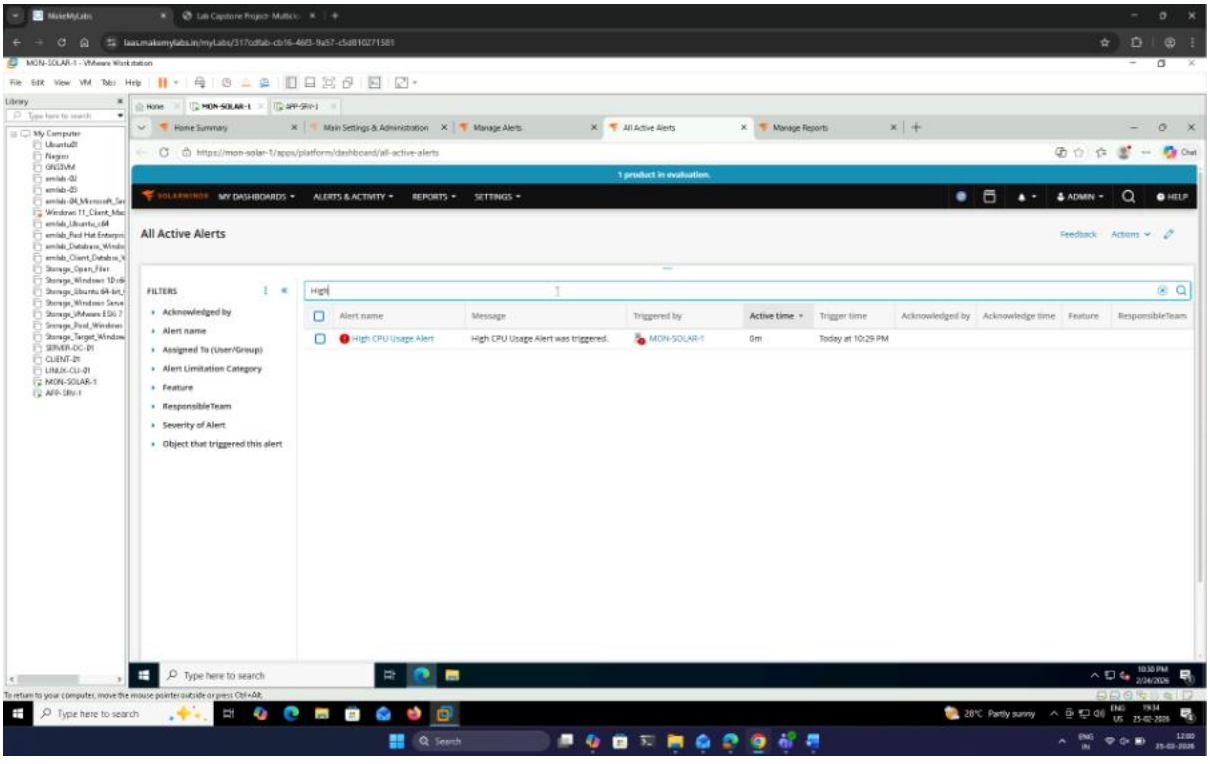





Home Summary
Actions
OVERVIEW
TOP 10
EVENTS
Top 10 Latency/Response Time
Node Name
IP Address
Vendor
Response Time
APP-SRV-1
192.168.159.160
Windows
11 ms
MON-SOLAR-1
192.168.159.153
Windows
10 ms
Top 10 Packet Loss
Node Name
IPAddress
Vendor
PercentLoss
MON-SOLAR-1
192.168.159.153
Windows
0 %
APP-SRV-1
192.168.159.160
Windows
0 %
Top 10 Interface Utilization
Interface
Node
InPercentUtil
OutPercentUtil
Intel(R) 82574L Gigabit Network...
APP-SRV-1
0 %
0 %
Local Netflow Source
MON-SOLAR-1
0 %
0 %
Top 10 CPU Utilization
Node Name
IPAddress
Vendor
CPU Usage
MON-SOLAR-1
192.168.159.153
Windows
61 %

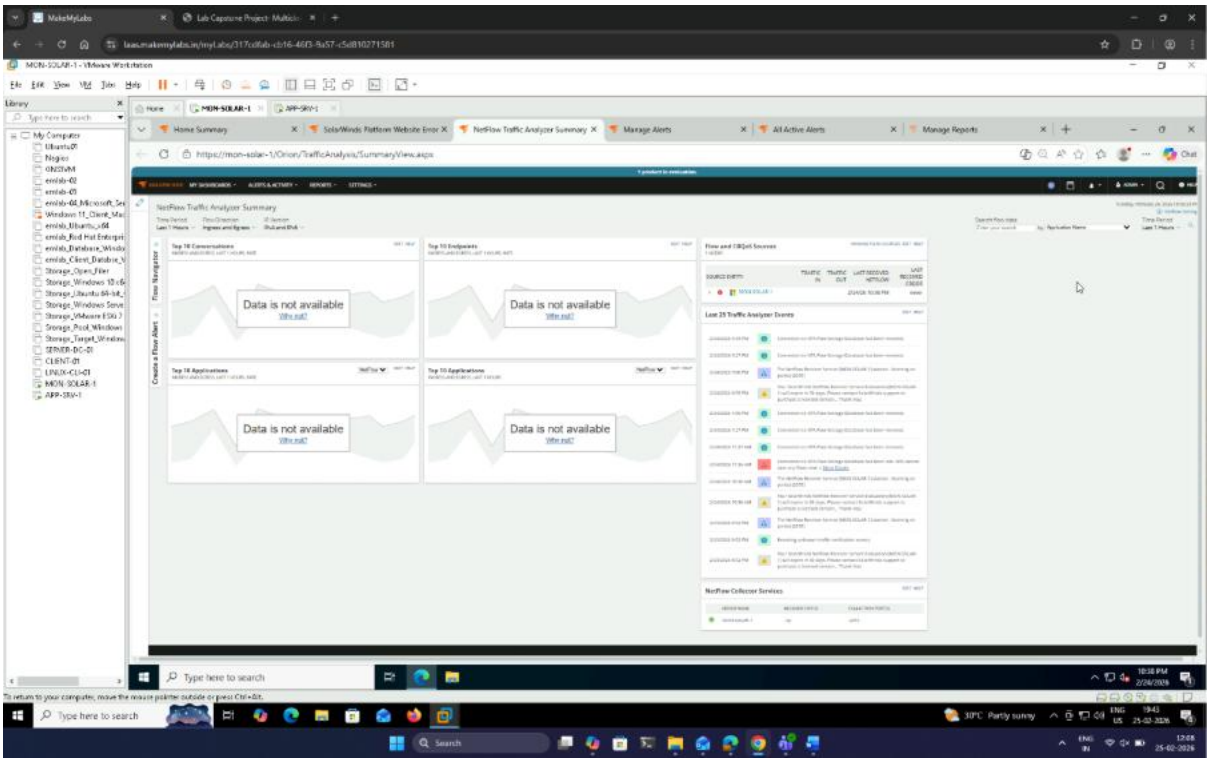

Home Summary
Actions
OVERVIEW
TOP 10
EVENTS
Intel(R) 82574L Gigabit Network...
APP-SRV-1
0 %
0 %
Local Netflow Source
MON-SOLAR-1
0 %
0 %
MON-SOLAR-1
192.168.159.153
Windows
61 %
Top 10 Volume Utilization
Volume Name
Node Name
VolumePercentUsed
Space Available
C:\ Label: 6AA46581
APP-SRV-1
94.601 %
1.587 GB



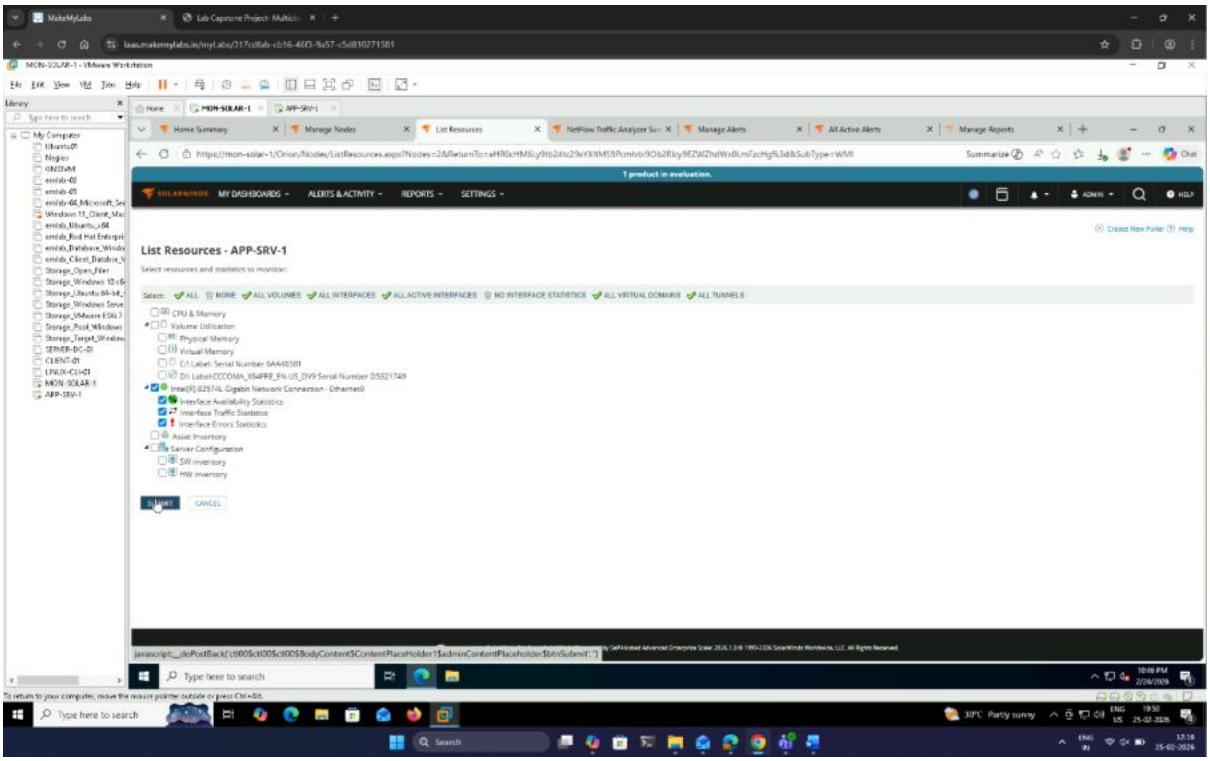


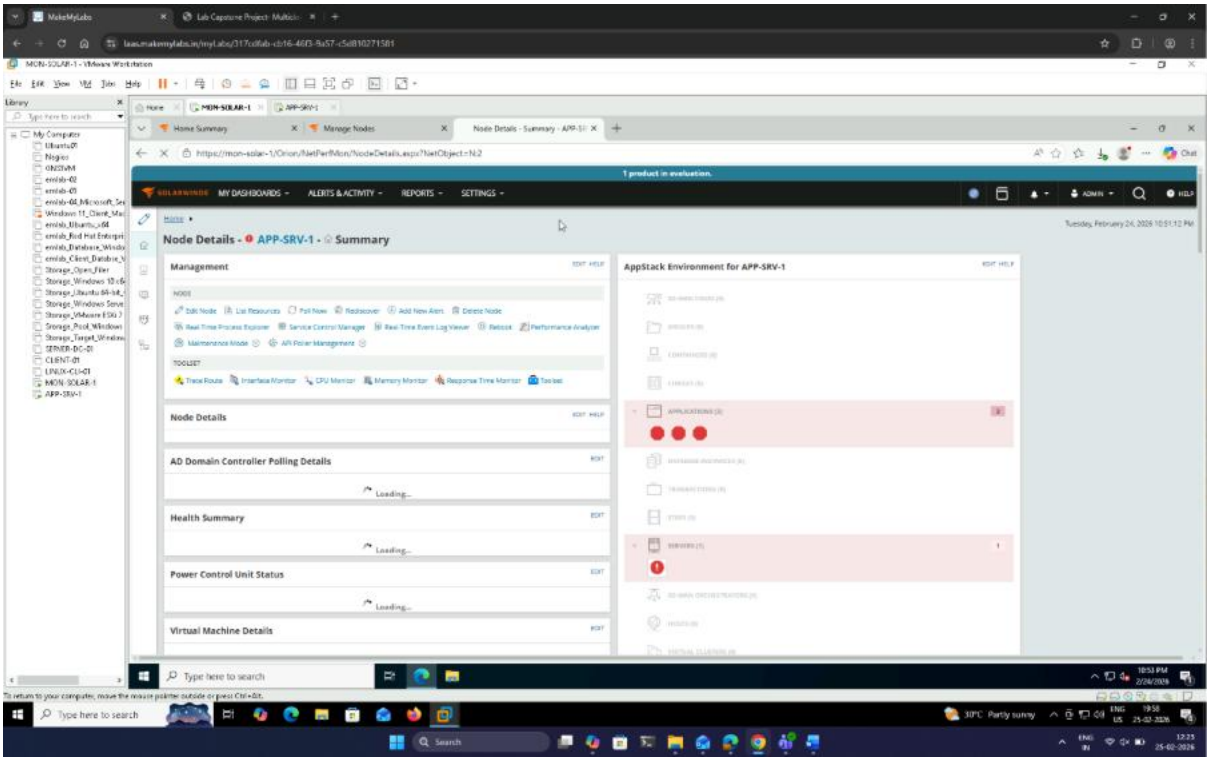
To validate the alert configuration, CPU utilization was artificially increased on the monitored system. SolarWinds successfully detected the threshold breach and generated

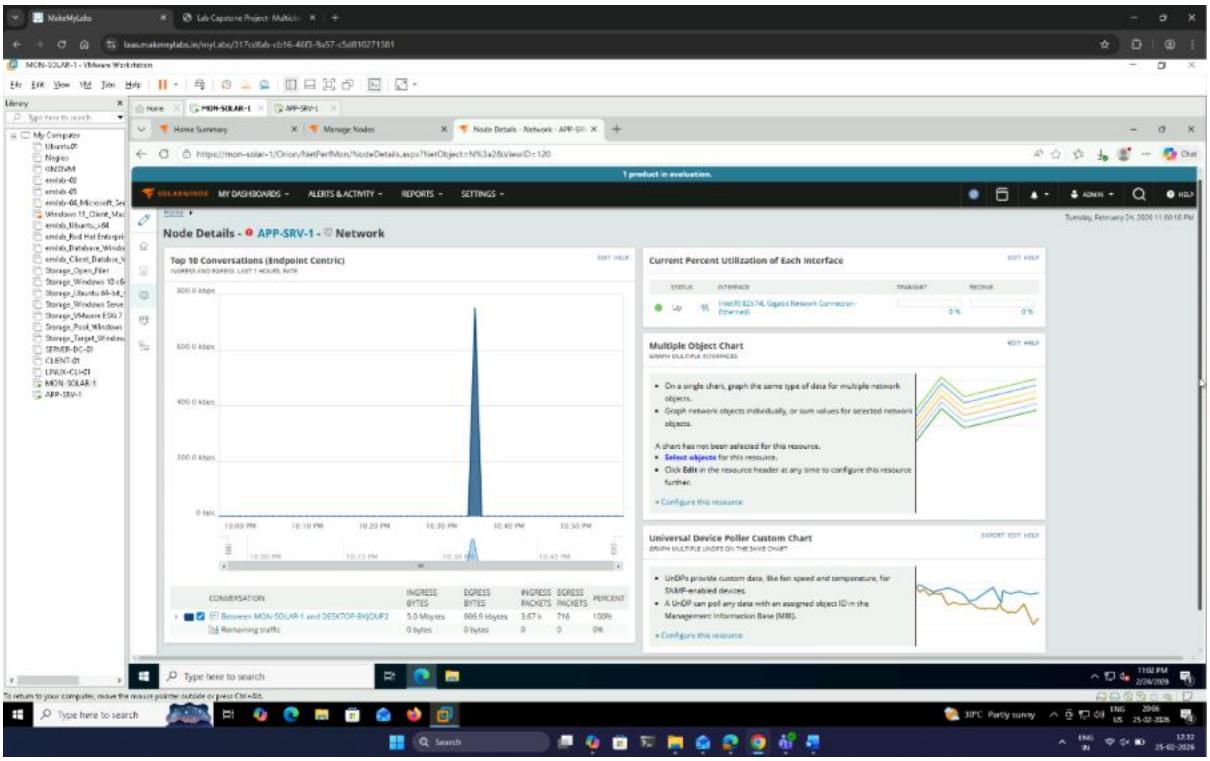
an active alert in the dashboard, confirming proper alert configuration and notification functionality.



We are getting the above output because NTA works only when the devices support NetFlow and Flow Exported in configured on router or switch.


List Resources - APP-SRV-1
Select resources and statistics to monitor:
SUBMIT
CANCEL


Node Details - APP-SRV-1 - Summary
Management
AppStack Environment for APP-SRV-1
Node Details
AD Domain Controller Polling Details
Health Summary
Power Control Unit Status
Virtual Machine Details



After successfully adding the node to SolarWinds, interface-level monitoring was configured to analyse network traffic statistics. The monitored node (APP-SRV-1) was accessed through the SolarWinds dashboard, and the “List Resources” option was used to verify and enable monitoring of network interfaces.

The primary network interface of the virtual machine was selected for monitoring. This enables SolarWinds to collect real-time statistics such as inbound traffic, outbound traffic, bandwidth utilization percentage, packet loss, and response time.

Once enables, the interface performance data was displayed in graphical format within the SolarWinds dashboard. These graphs provided visibility into network usage trends and allowed analysis of traffic patterns over time.

This step fulfilled the traffic analysis requirement of the monitoring assignment without requiring NetFlow configuration, as interface-based monitoring is sufficient in a virtual lab environment.

Nagios

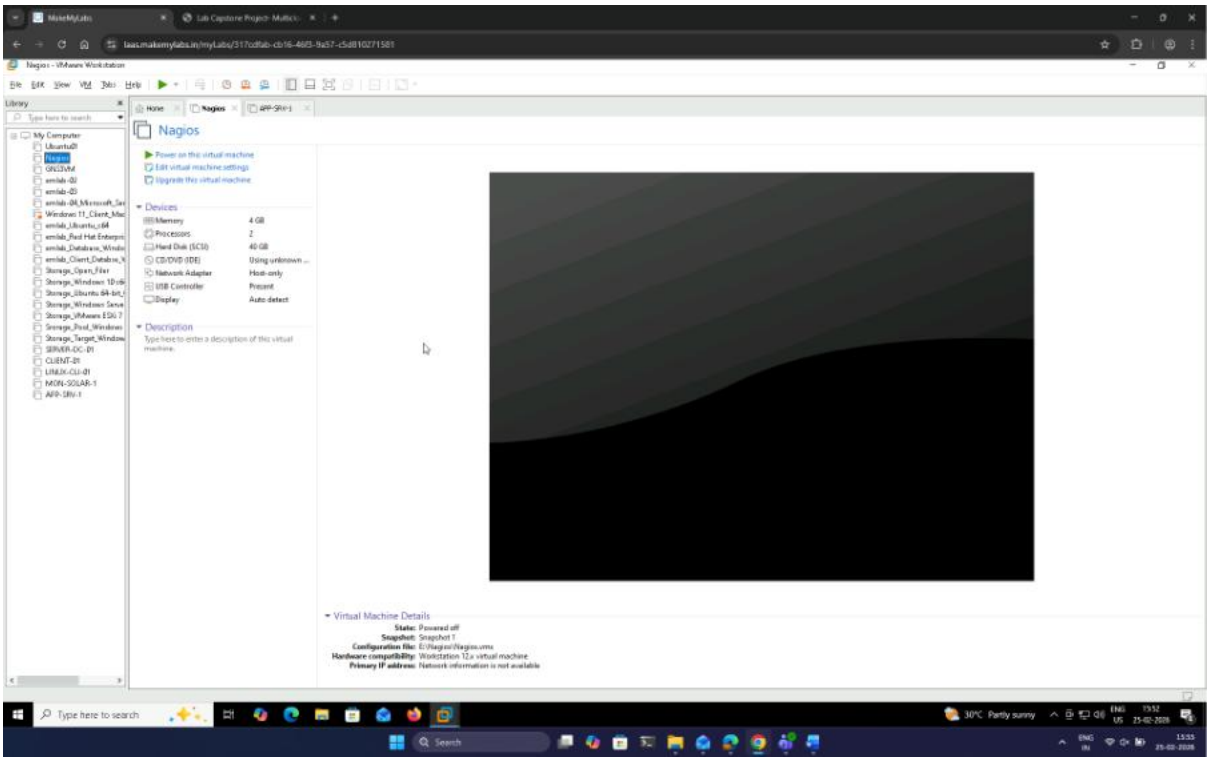
- Installation and Initial Setup:
    - Install Nagios on a virtual machine or server.
    - Configure the initial setup and connect it to a sample network.

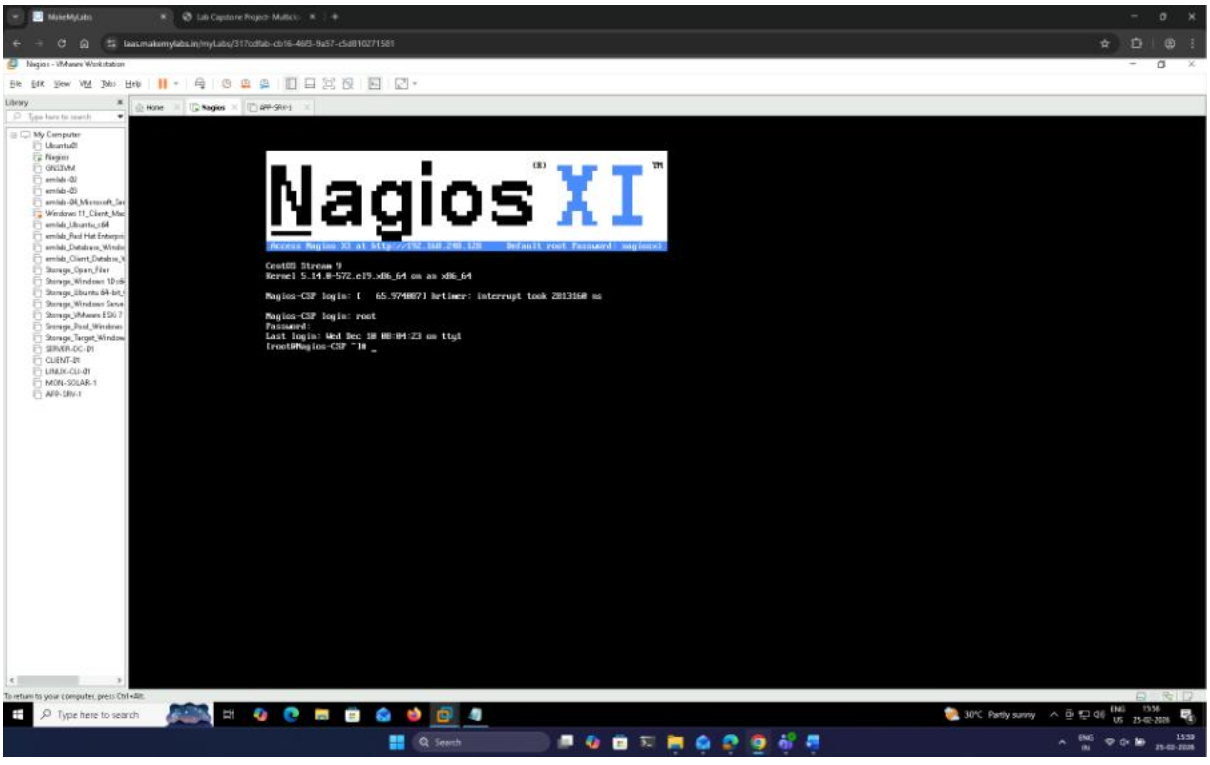
- Agent Deployment:
    - Deploy Nagios agents to various devices.

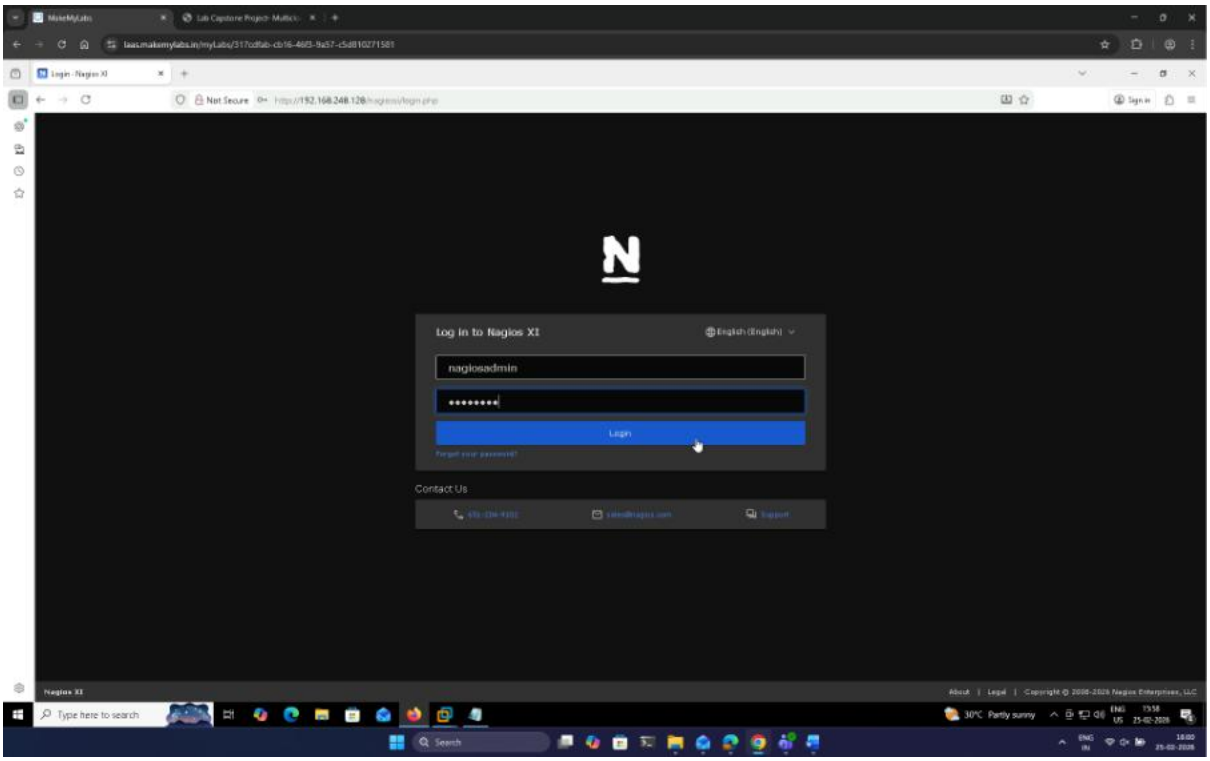
- Management Packs:
    - Import and configure management packs for different services.
    - Create custom management packs for specific monitoring needs.

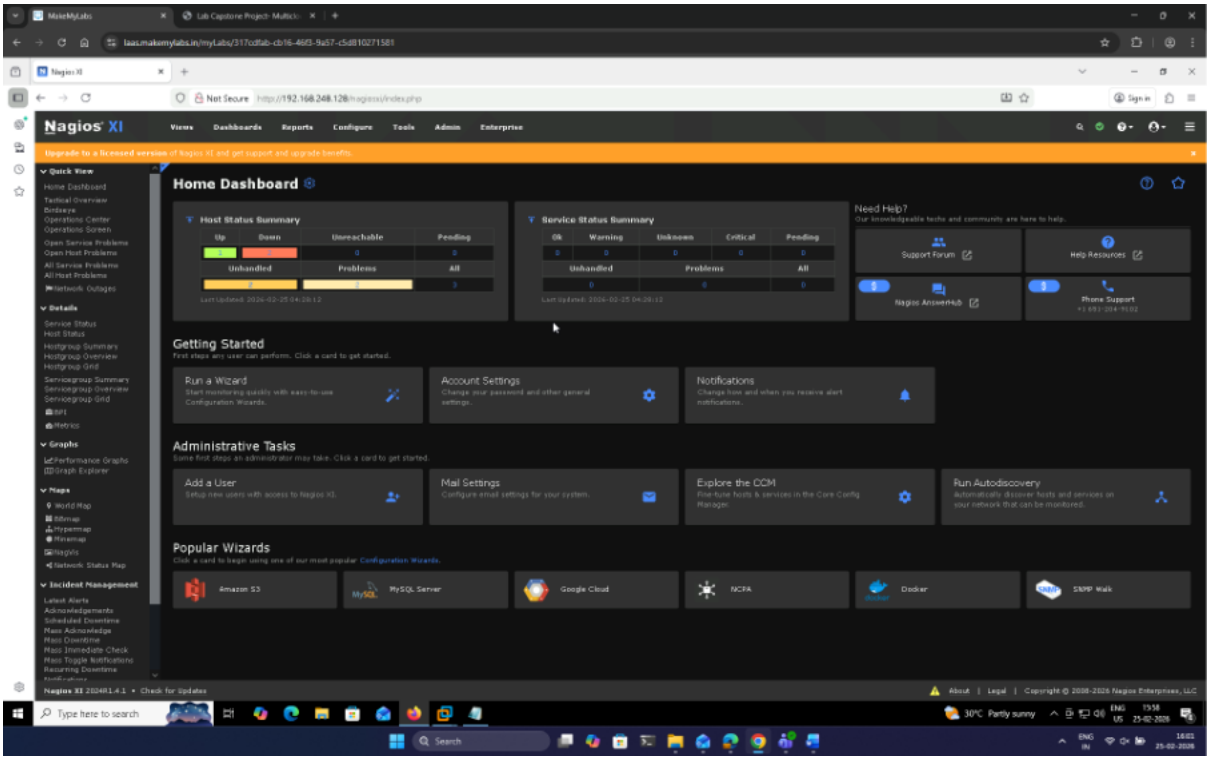
- Notifications and Reporting:
    - Configure notifications for different events
    - Set up reporting and service level tracking

Nagios was selected as an alternative monitoring solution due to its open-source architecture and lightweight resource requirements. Unlike SCOM, which requires additional enterprise components and licensing, Nagios can be deployed efficiently in a virtual lab environment. A dedicated Virtual Machine Nagios was created to simulate centralized infrastructure monitoring. This approach ensures isolation between monitoring platforms and reflects real-world deployment practices.

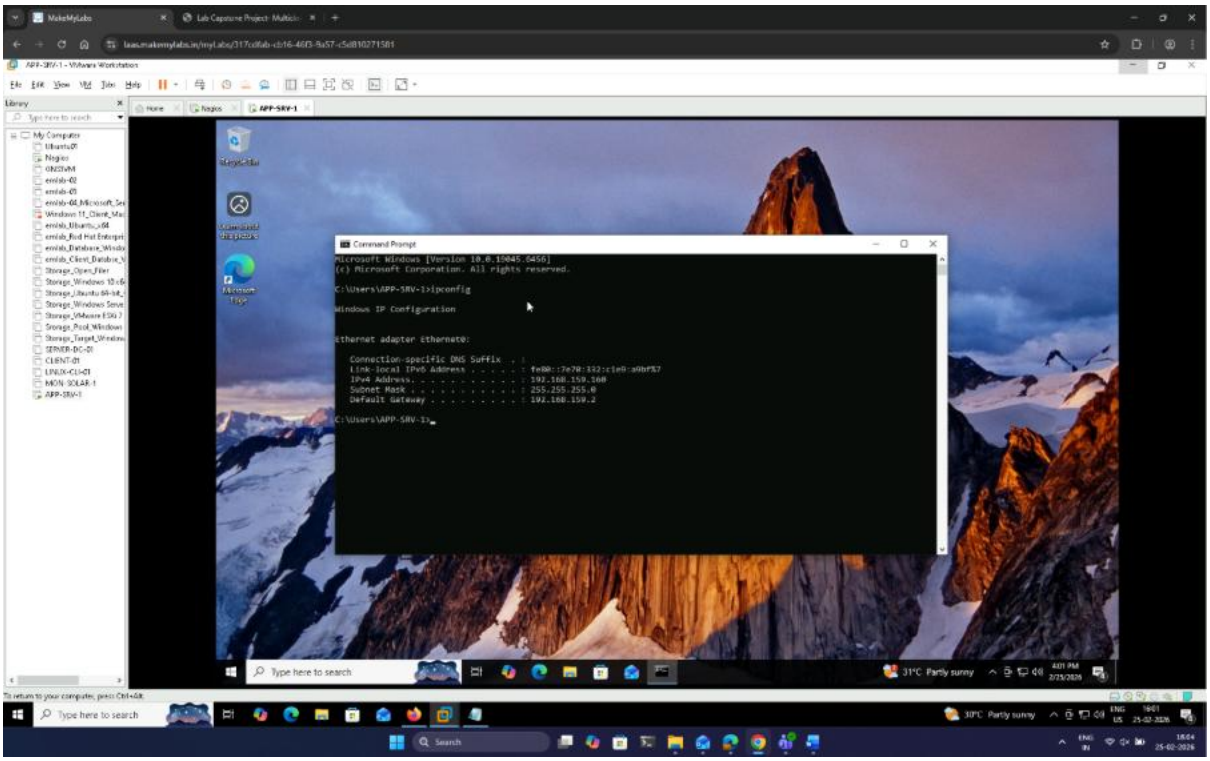

Nagios
Power on this virtual machine
Edit virtual machine settings
Upgrade this virtual machine
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 40 GB
CD/DVD (IDE) Using unknown ...
Network Adapter Host-only
USB Controller Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Snapshot: Snapshot 1
Configuration file: E:\Nagios\Nagios.vmx
Hardware compatibility: Workstation 12.x virtual machine
Primary IP address: Network information is not available


Nagios XI
CentOS Stream 9
Kernel 5.14.0-572.el9.x86_64 on an x86_64
Nagios-CSP login:
Password:
Last login: Wed Dec 10 08:04:23 on tty1
[root@Nagios-CSP ~]#

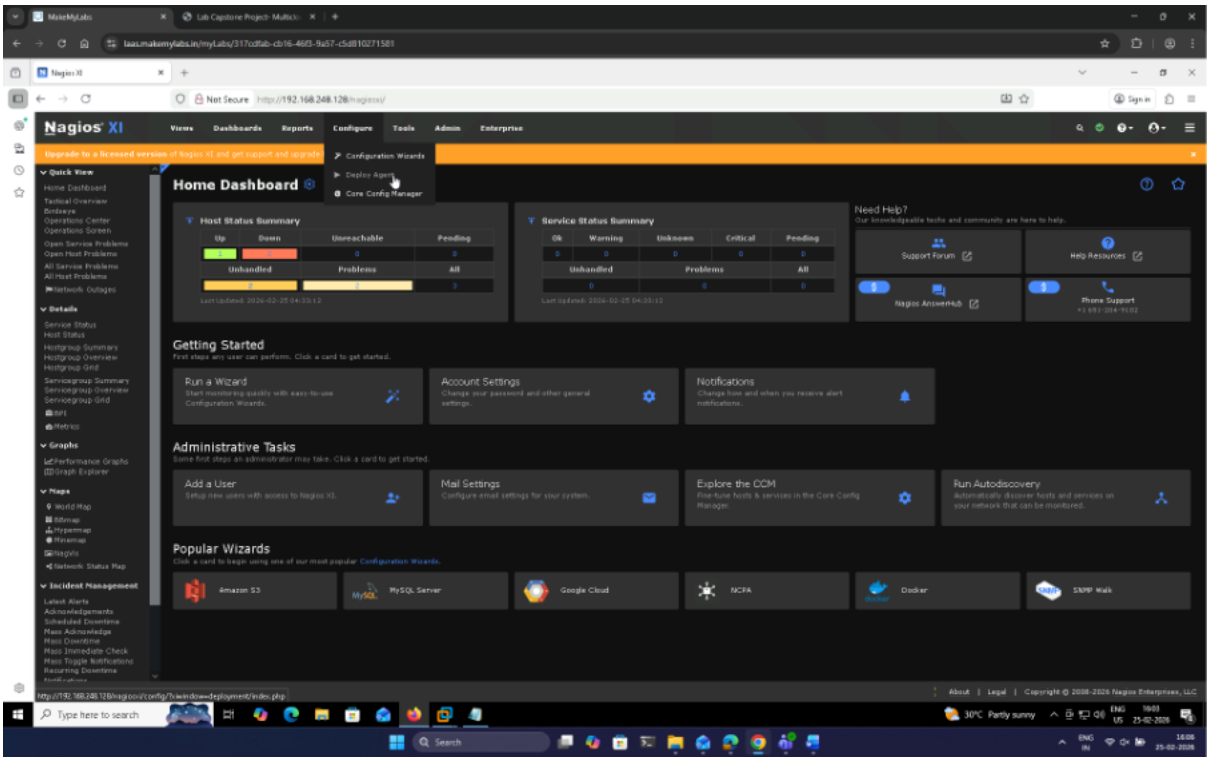


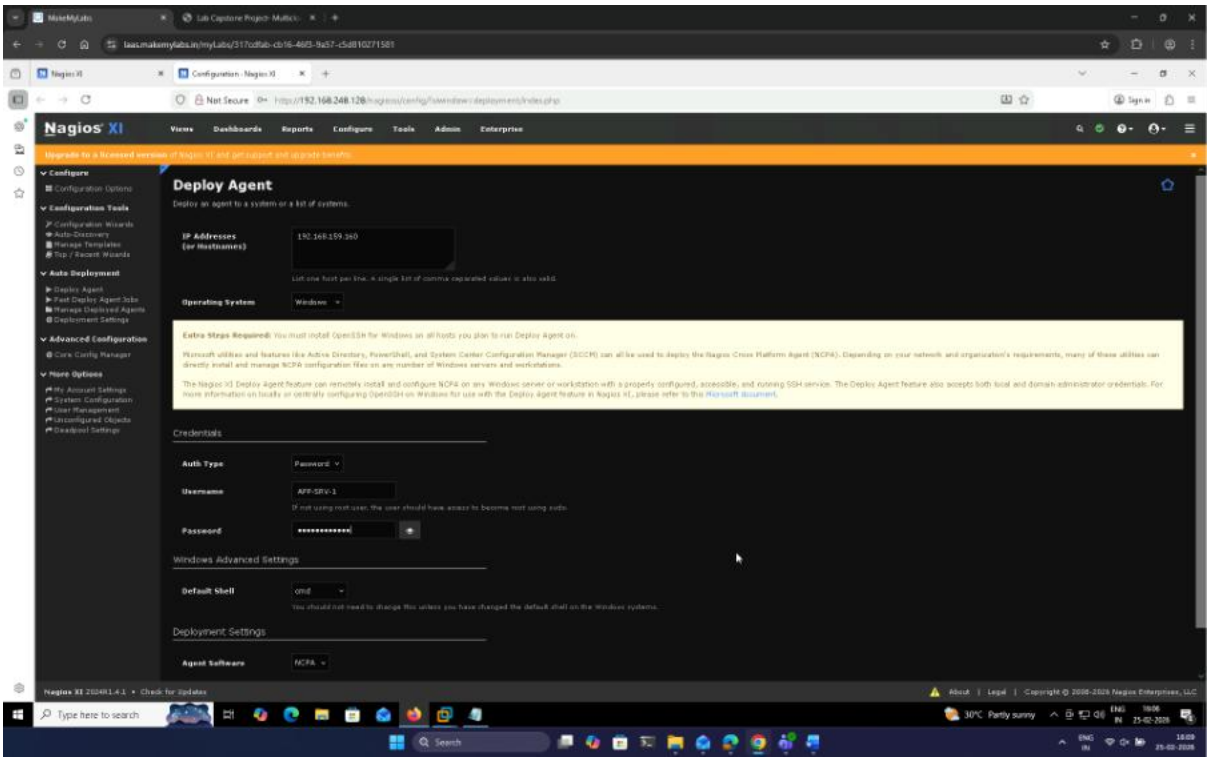


The Nagios machine has been powered on and logged into the web console.

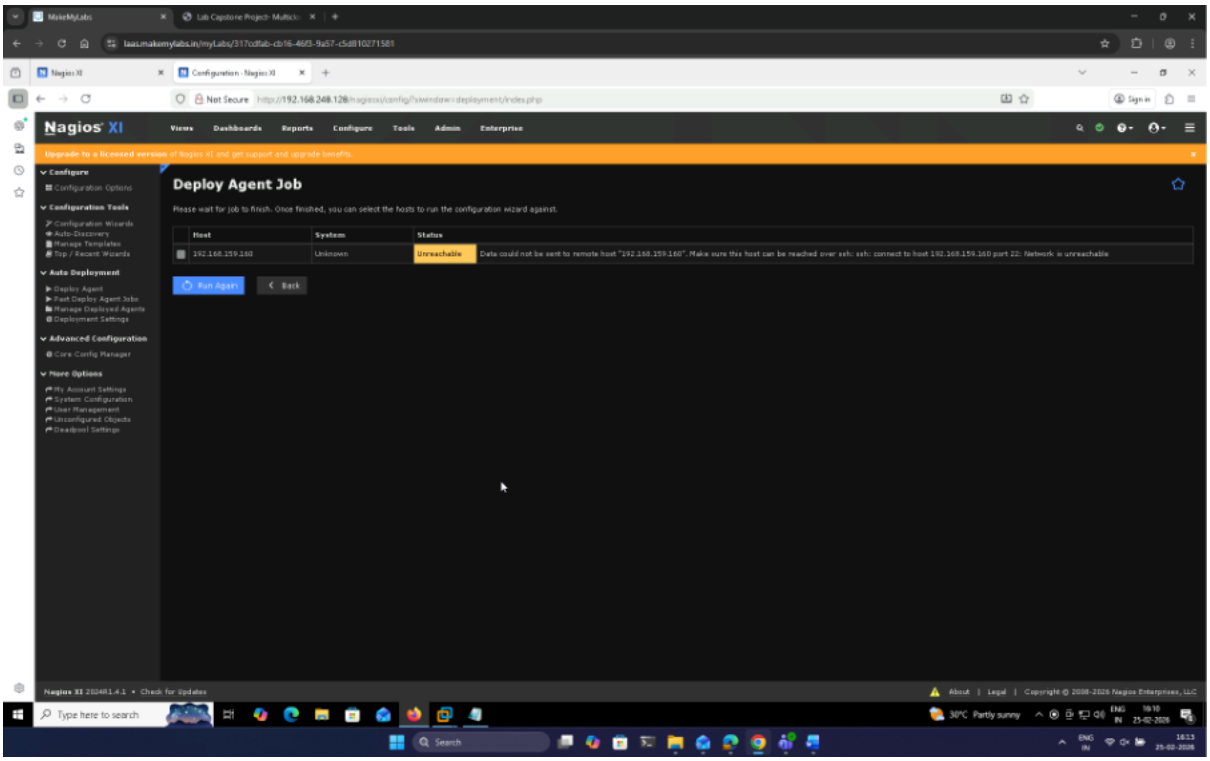


We have note down the IP address of APP-SRV-1 machine which is 192.168.159.160 so that we can add it into the nagios as a monitored host.



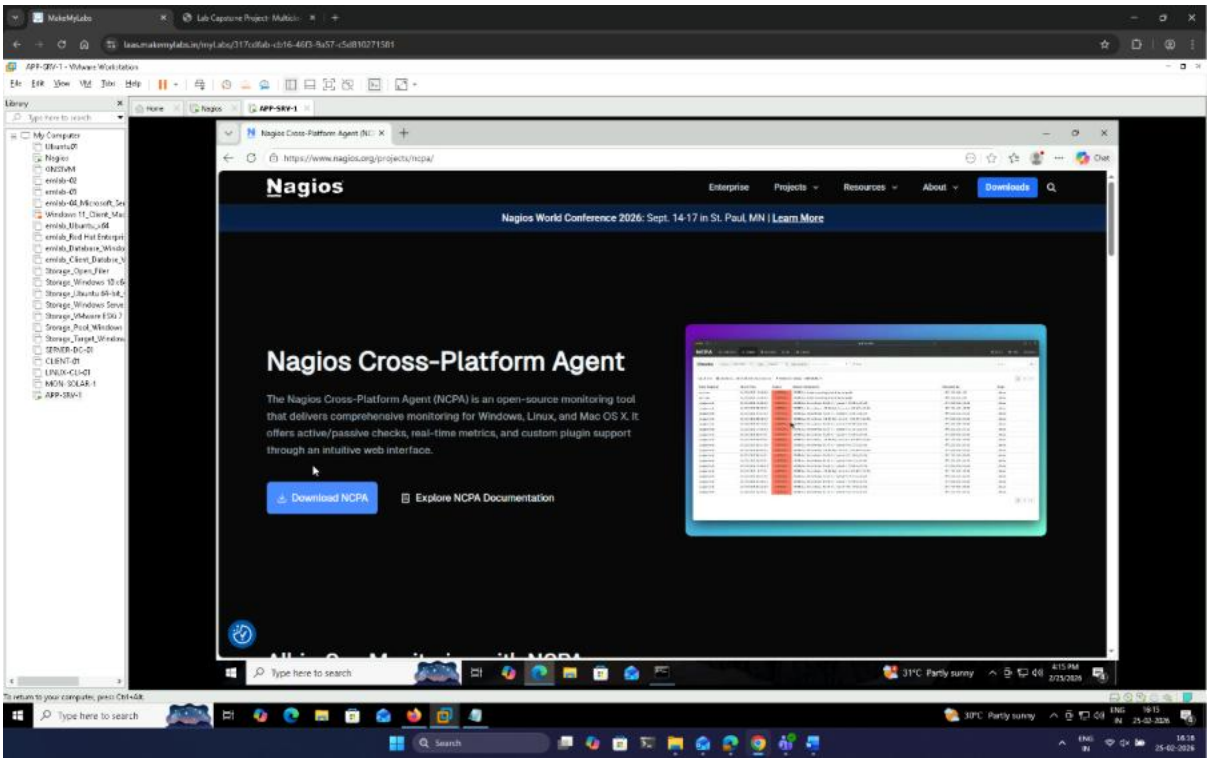


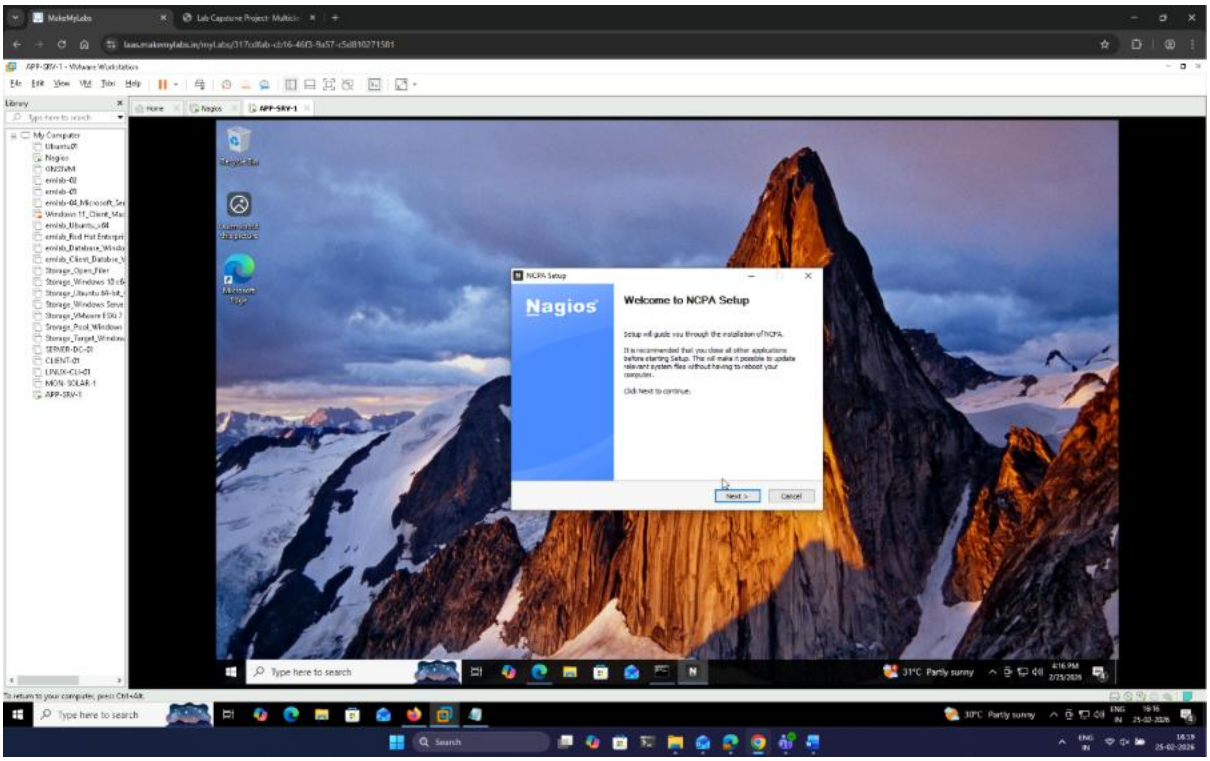
Nagios XI was used to deploy the NCPA (Nagios Cross Platform Agent) to the Windows Client Machine APP-SRV-1 to enable advanced monitoring capabilities.

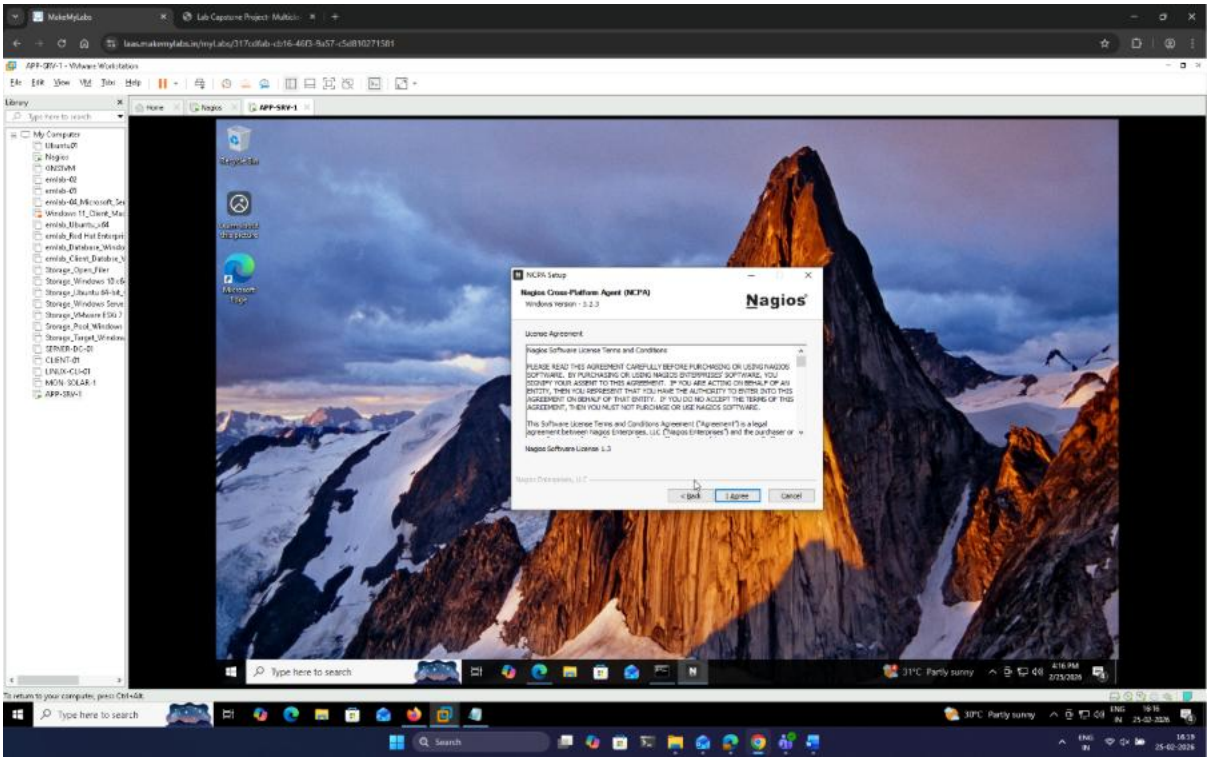


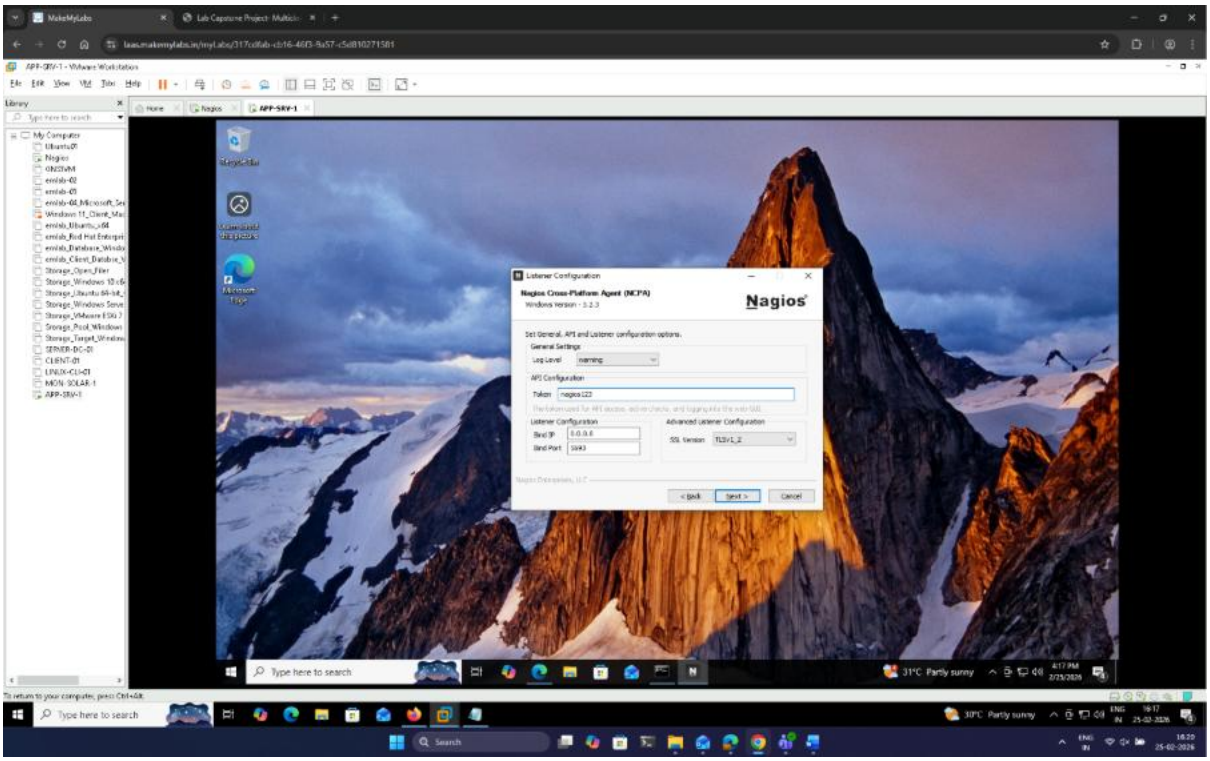
Nagios is trying to connect using SSH (Port 22) but SSH is used in Linux and Windows 10 does not use SSH by default so what we will do now is install NCPA manually on APP-SRV-1 and then configure Nagios to use the NCPA API.

We will download the installer from the link ***(https://www.nagios.org/ncpa/)***.

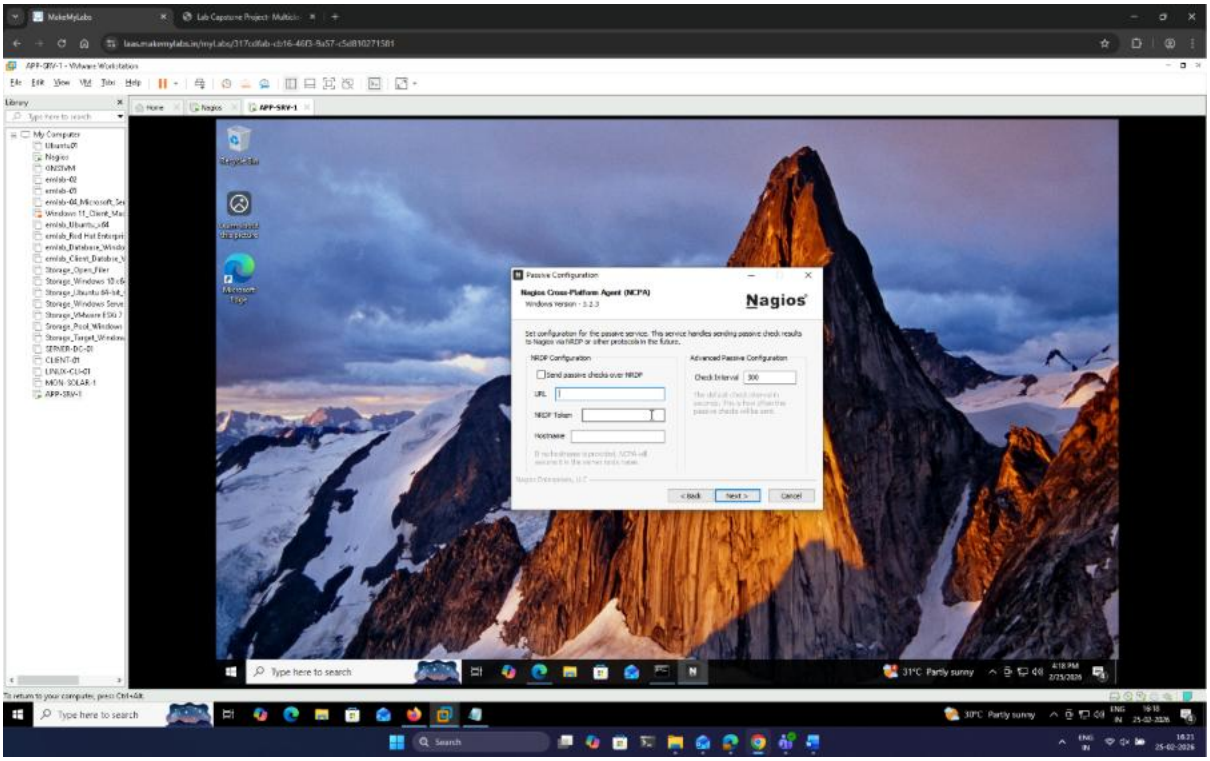



NCPA Setup
Nagios
Welcome to NCPA Setup
Setup will guide you through the installation of NCPA.
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Next >
Cancel


NCPA Setup
Nagios Cross-Platform Agent (NCPA)
Nagios
License Agreement
I Agree
Cancel

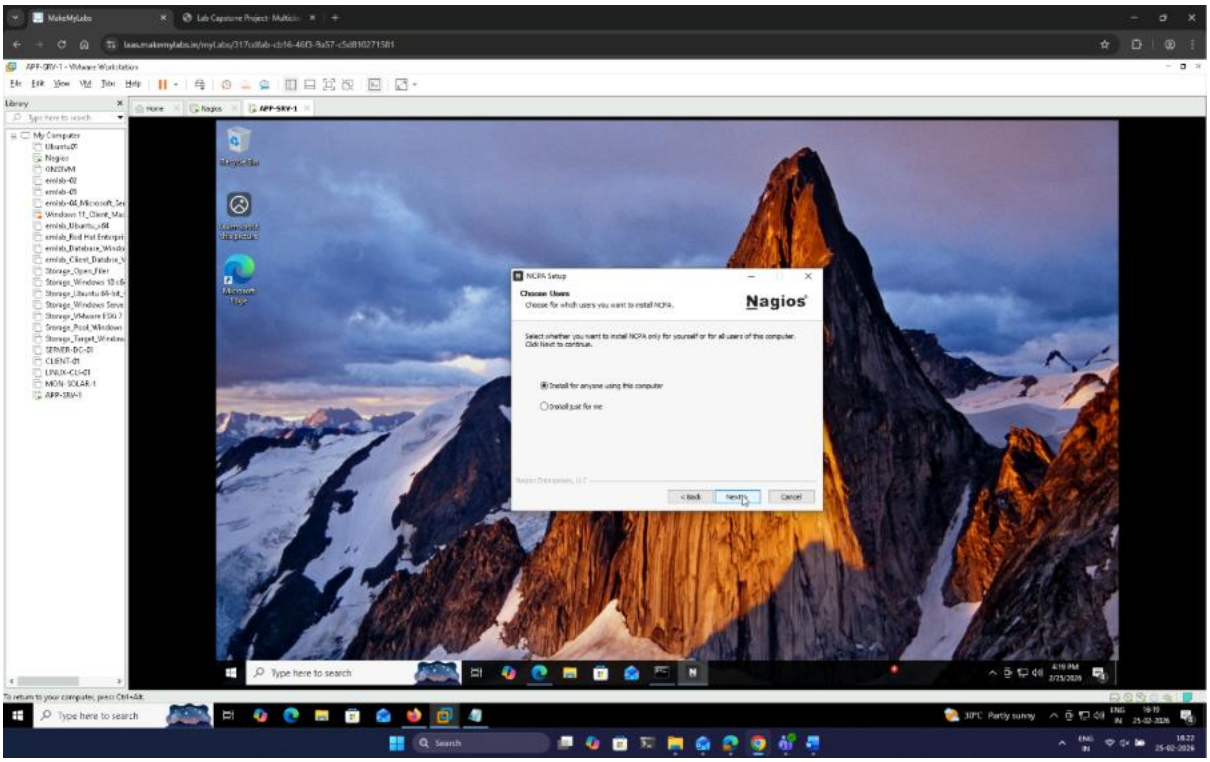


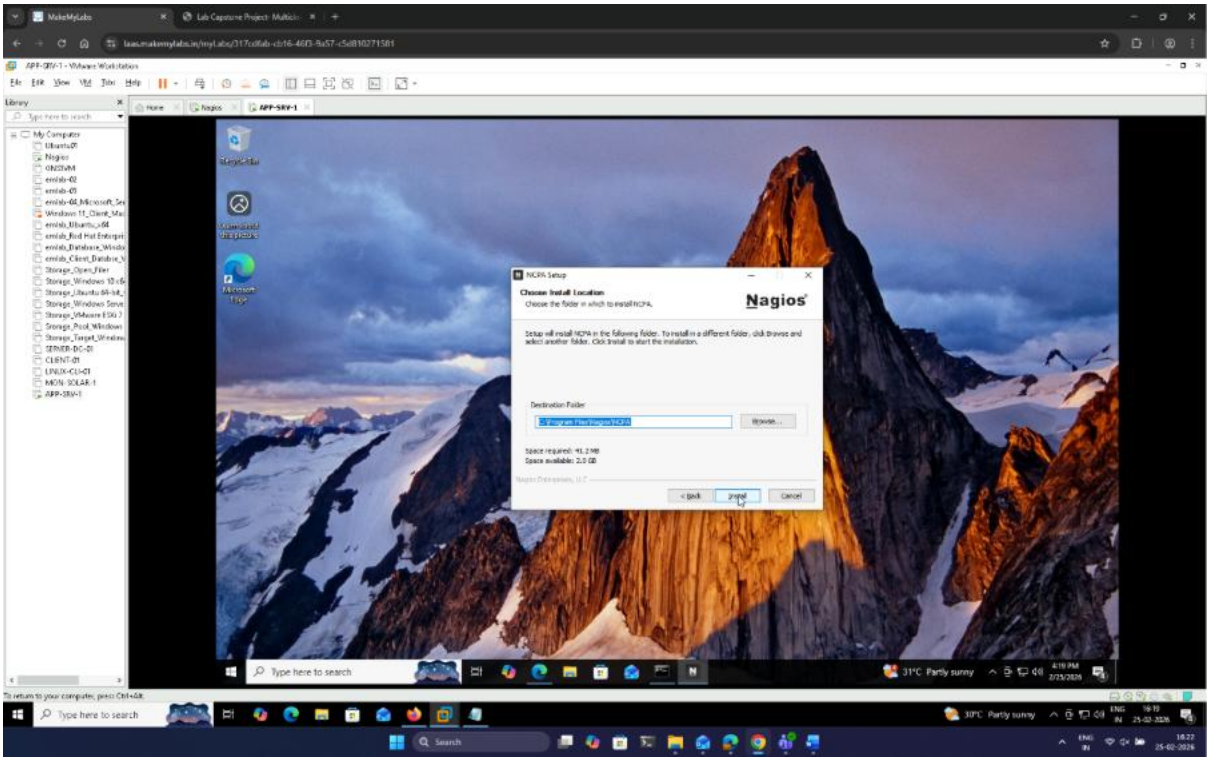
We have set out API key as: - ***nagios123***.

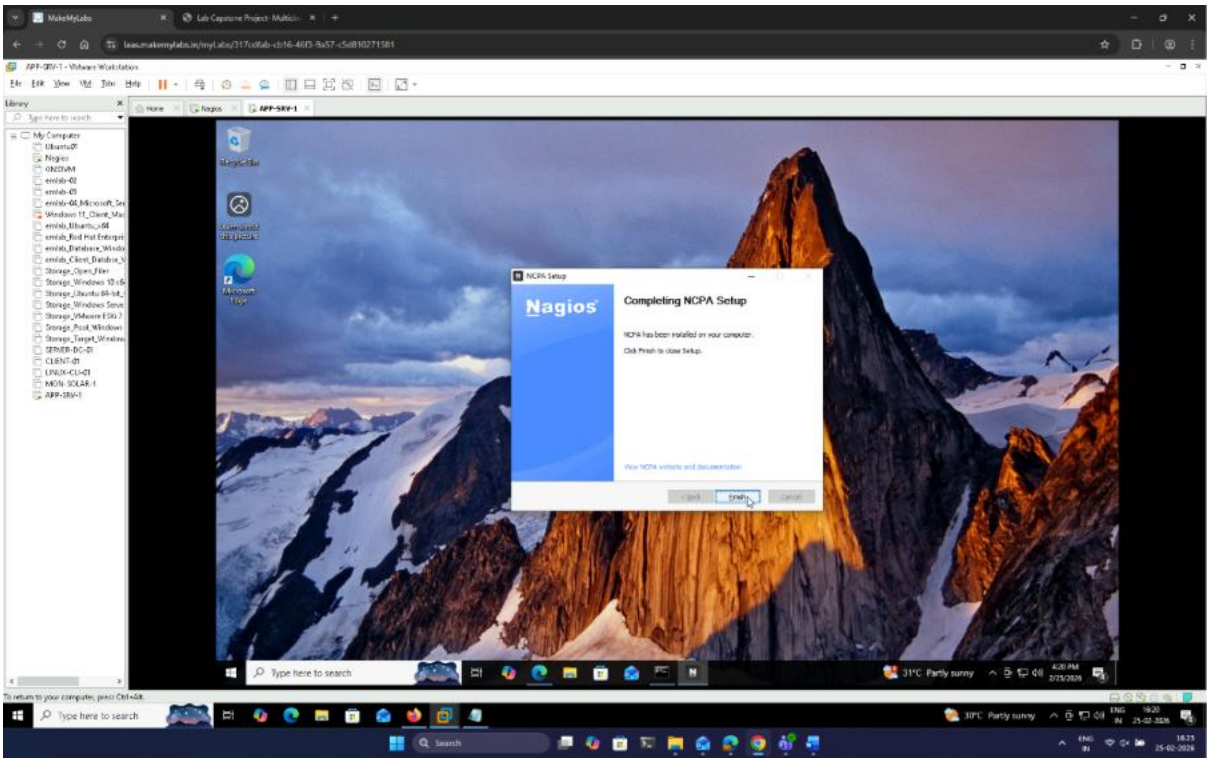


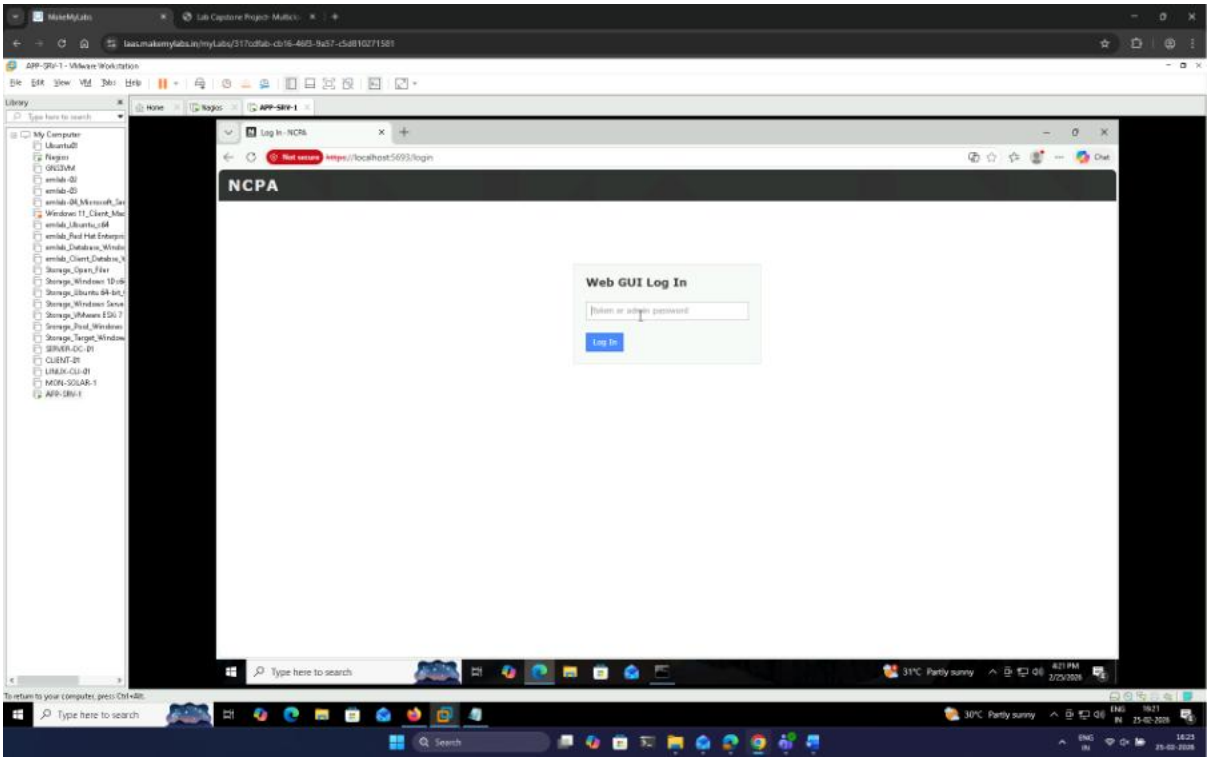
During the NCPA installation, the passive configuration (NRDP) option was left disabled as the monitoring setup uses active polling from the Nagios XI server. This ensures that the

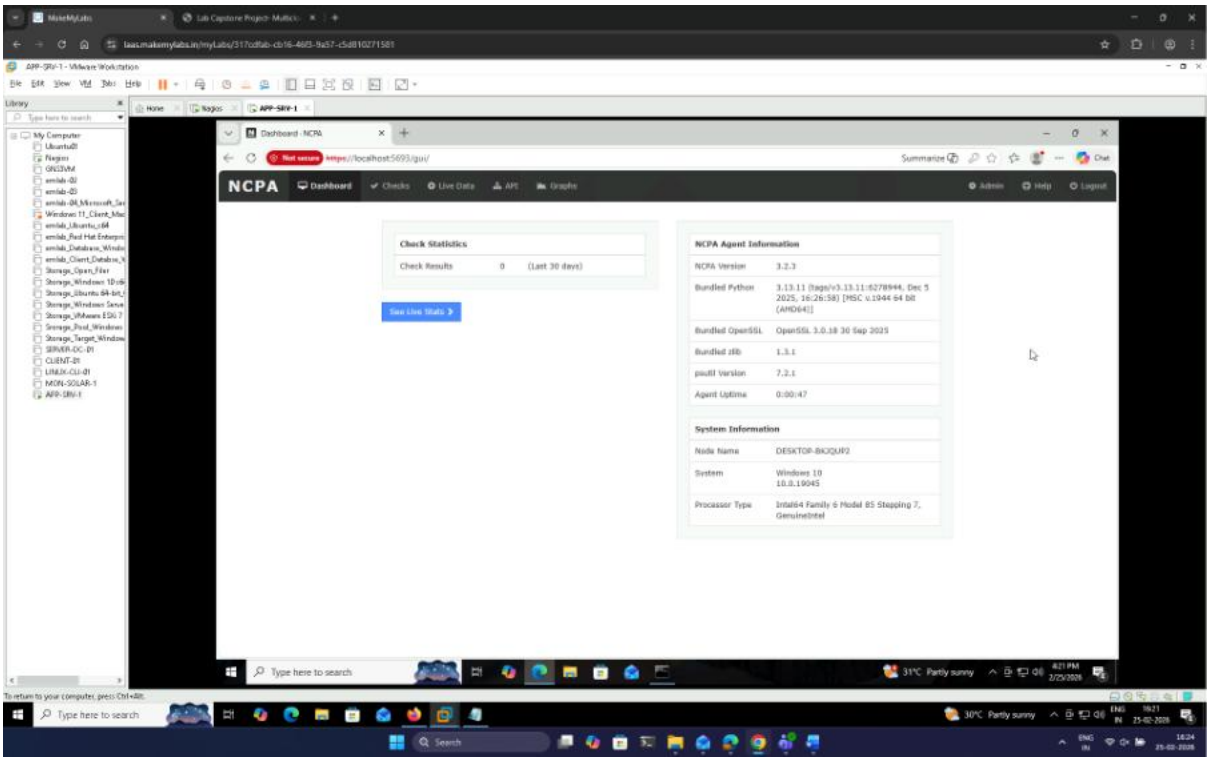
Nagios server initiates checks to retrieve system performance metrics from the Windows client through secure API communication.



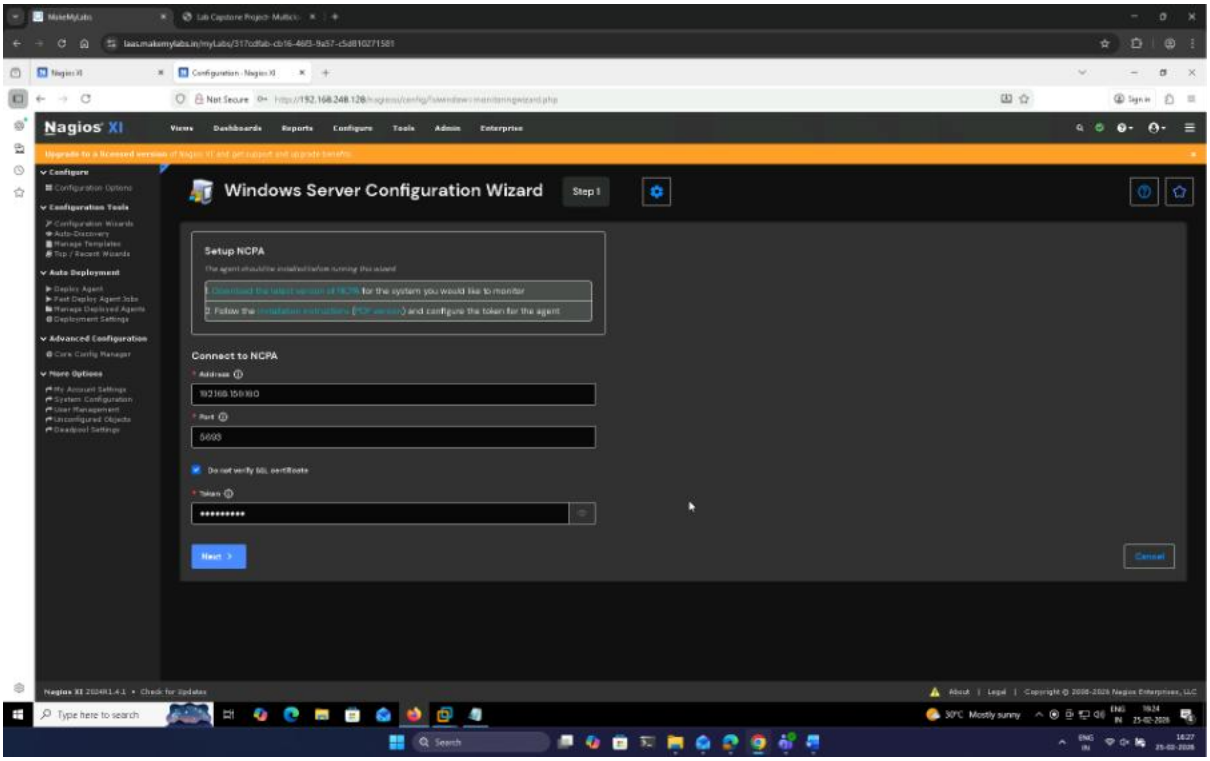


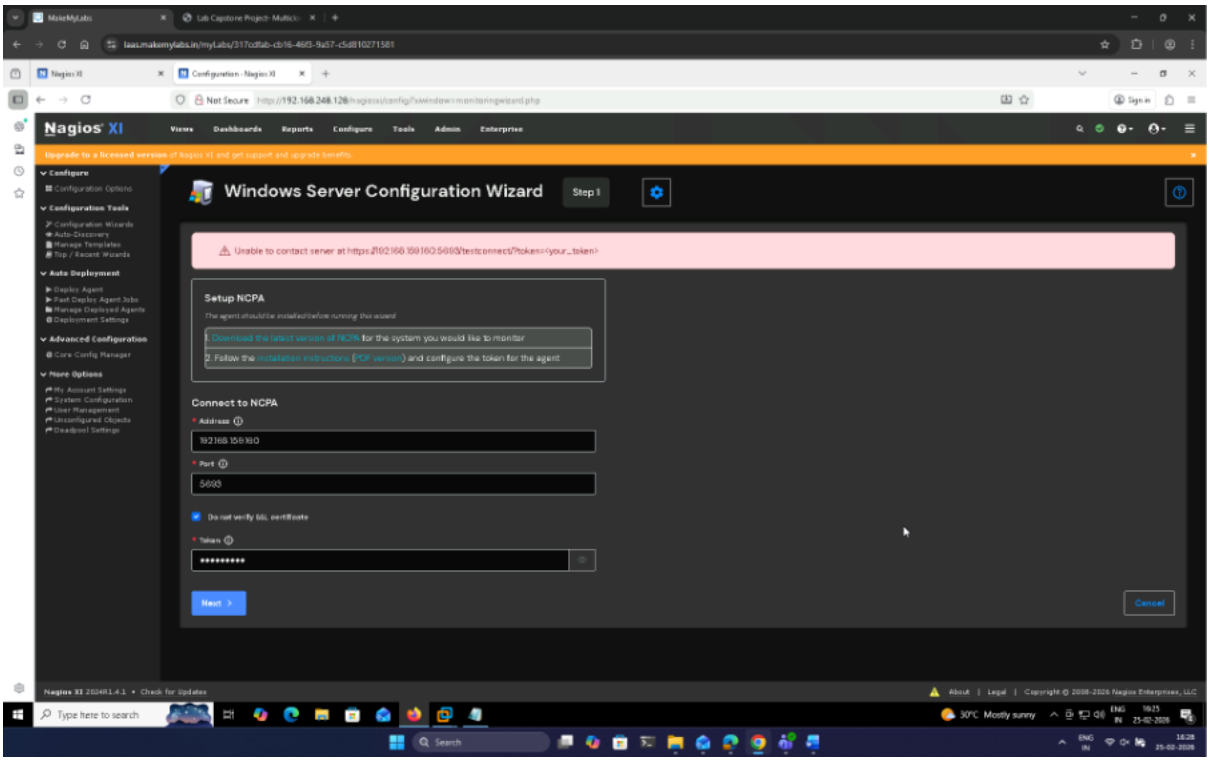

NCPA Setup
Nagios
Completing NCPA Setup
NCPA has been installed on your computer.
Click Finish to close Setup.
Finish


Log In - NCPA
https://localhost:5693/login
NCPA
Web GUI Log In
Token or admin password
Log In

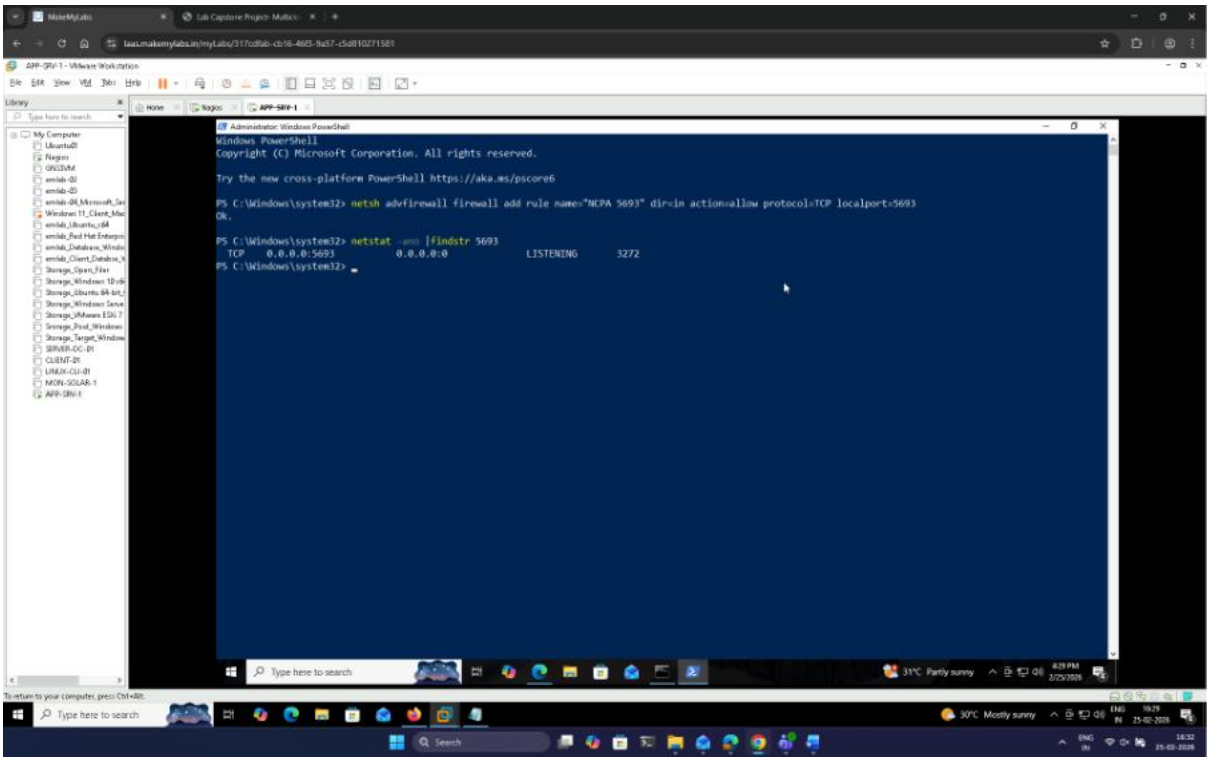


We have successfully setup the NCPA Client on Windows Machine (APP-SRV-1).

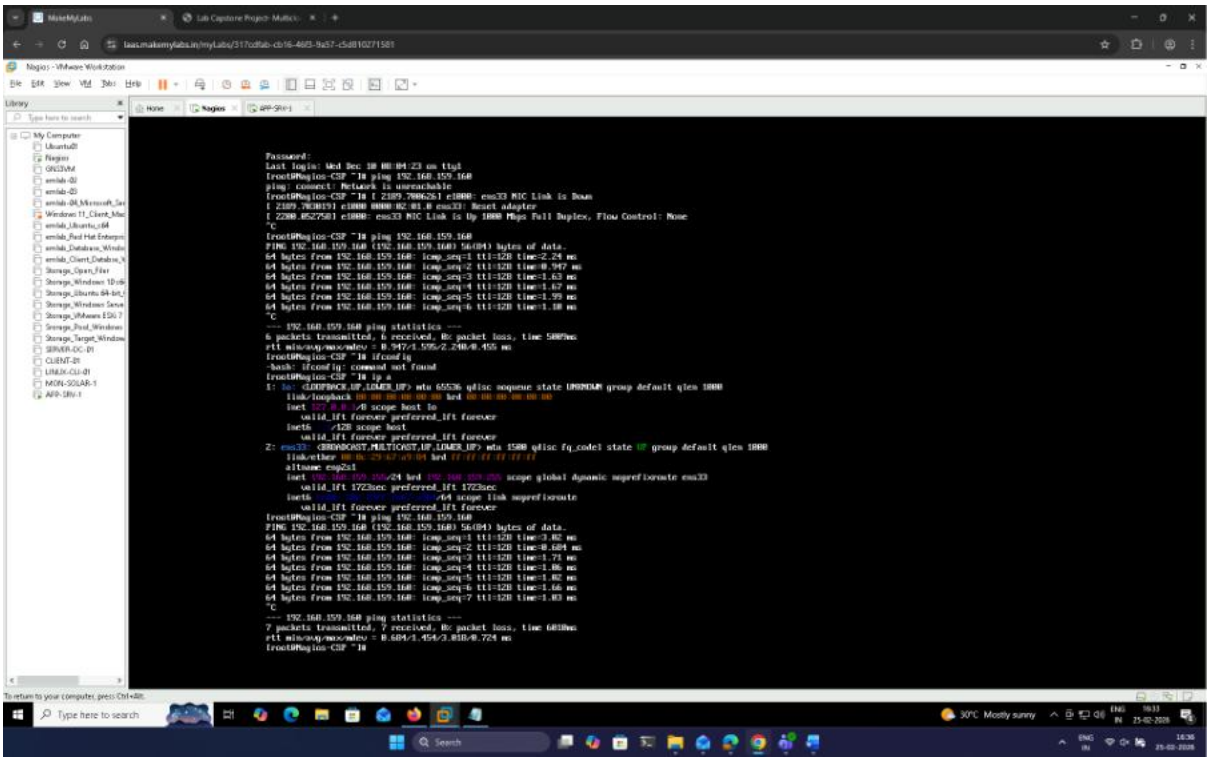




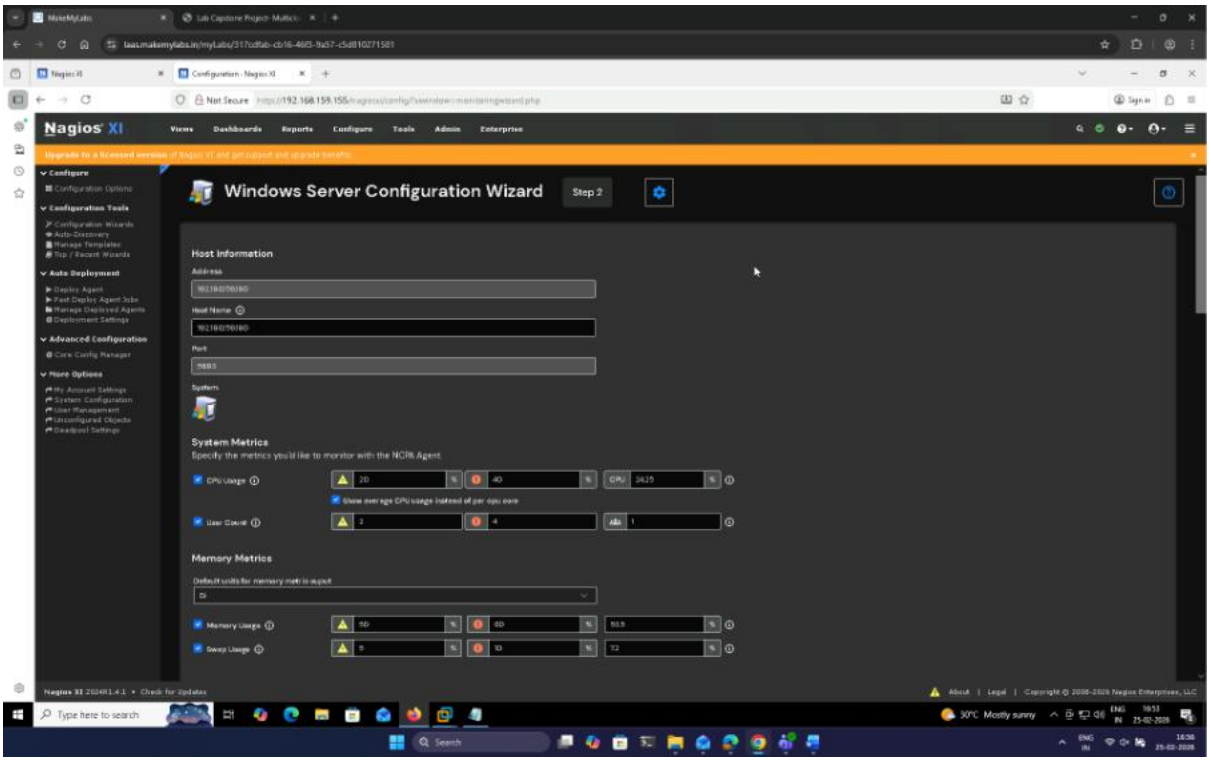
The problem ***Unable to connect at server at https://192.168.159.160:5693...*** means that Nagios cannot reach Windows 10 on port 5693 which is a Windows Firewall issue.

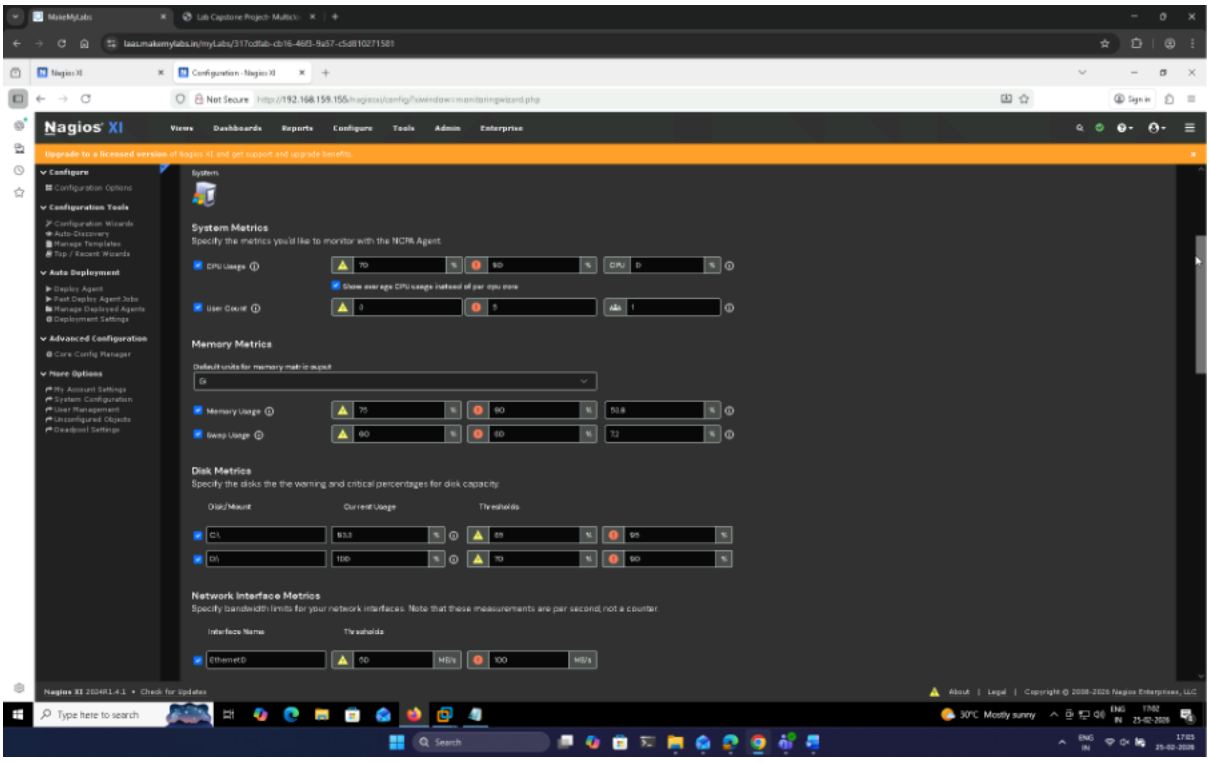


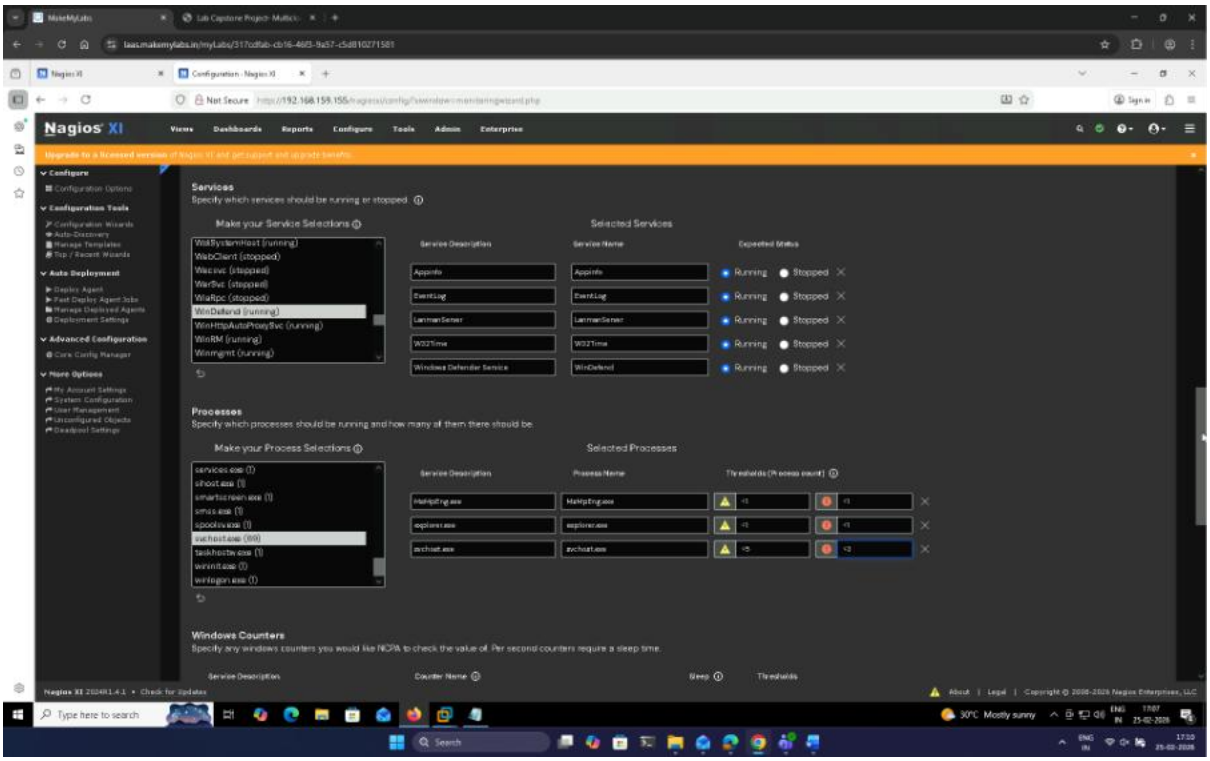
We have successfully added a firewall rule that allows communication on the port 5693.

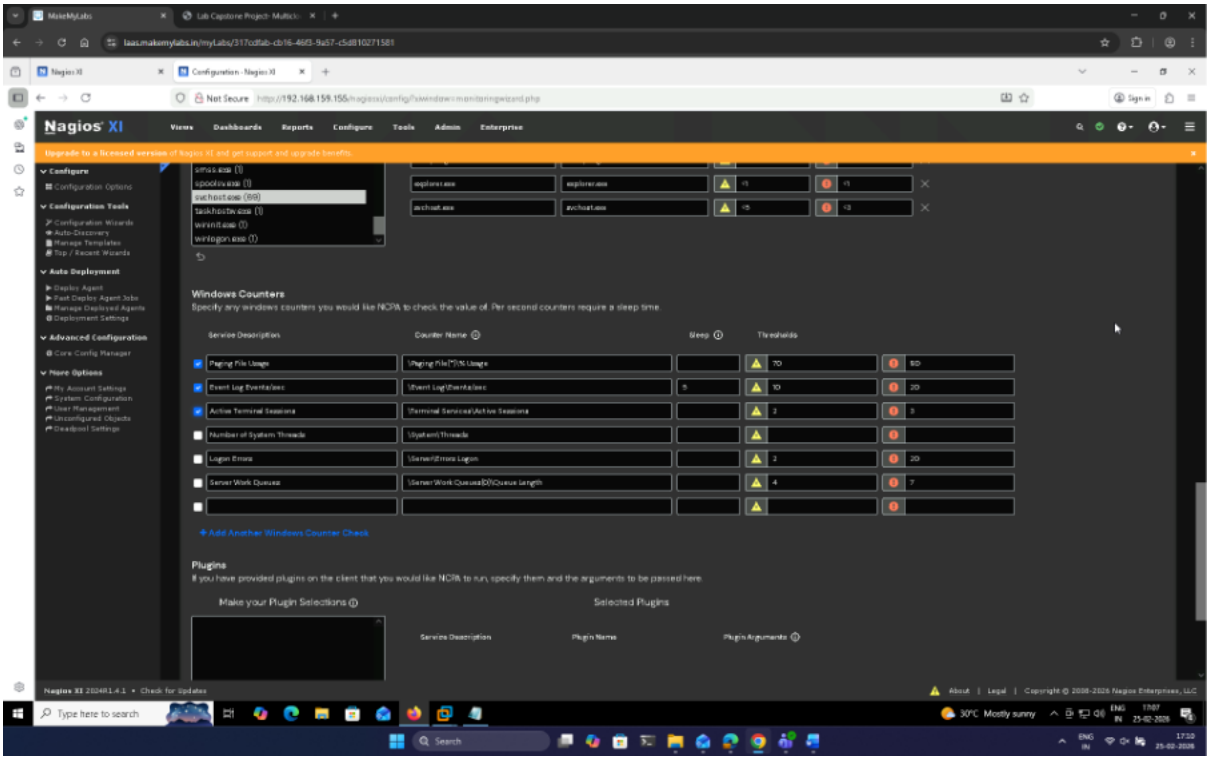


During the configuration of Nagios XI to monitor the Windows 10 Client, connectivity issues were encountered due to subnet mismatch between the Nagios server and the monitored host. The systems were initially configured on different virtual network segments, preventing communication over port 5693. The issue was resolved by aligning both virtual machines to the same network adapter configuration and allowing inbound firewall rules for the NCPA service.


Nagios XI
Windows Server Configuration Wizard
Step 2
Host Information
System Metrics
Specify the metrics you'd like to monitor with the NCPA Agent
CPU Usage
User Count
Memory Metrics
Memory Usage
Swap Usage


Nagios XI
System Metrics
Specify the metrics you'd like to monitor with the NCPA Agent
CPU Usage
User Count
Memory Metrics
Memory Usage
Swap Usage
Disk Metrics
Specify the disks the the warning and critical percentages for disk capacity
Network Interface Metrics
Specify bandwidth limits for your network interfaces. Note that these measurements are per second, not a counter



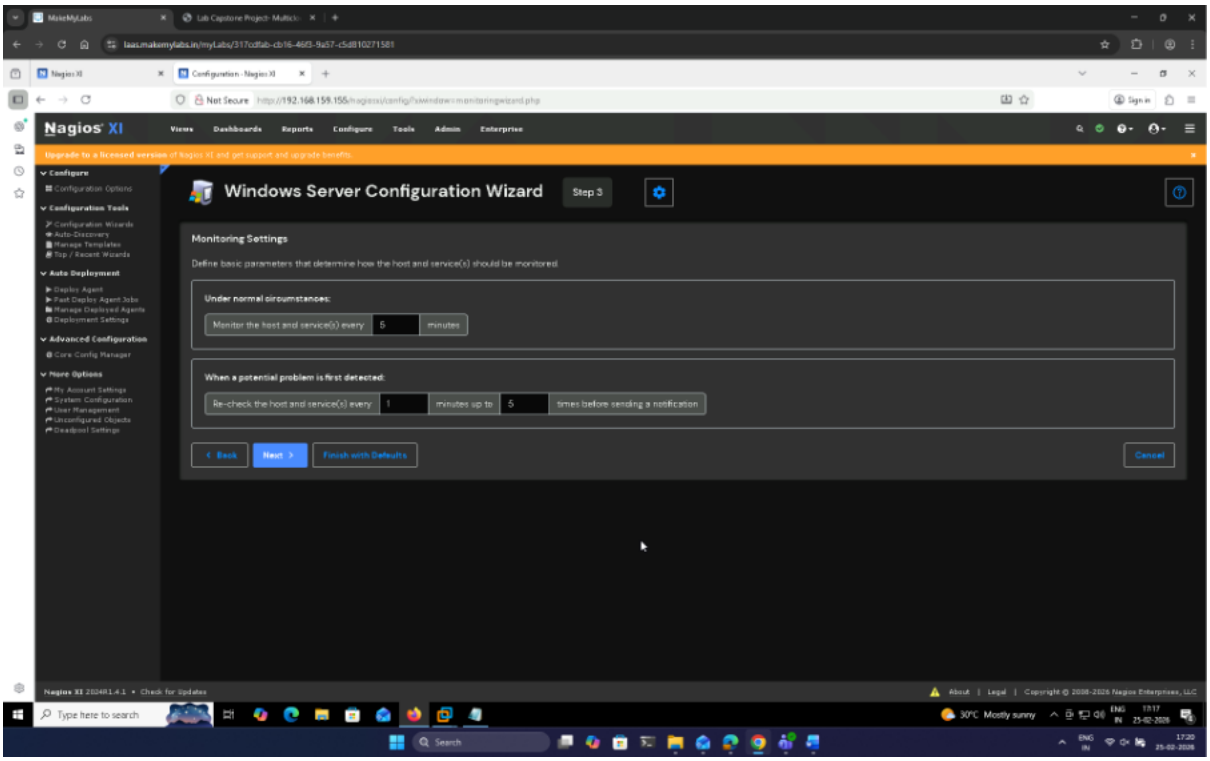


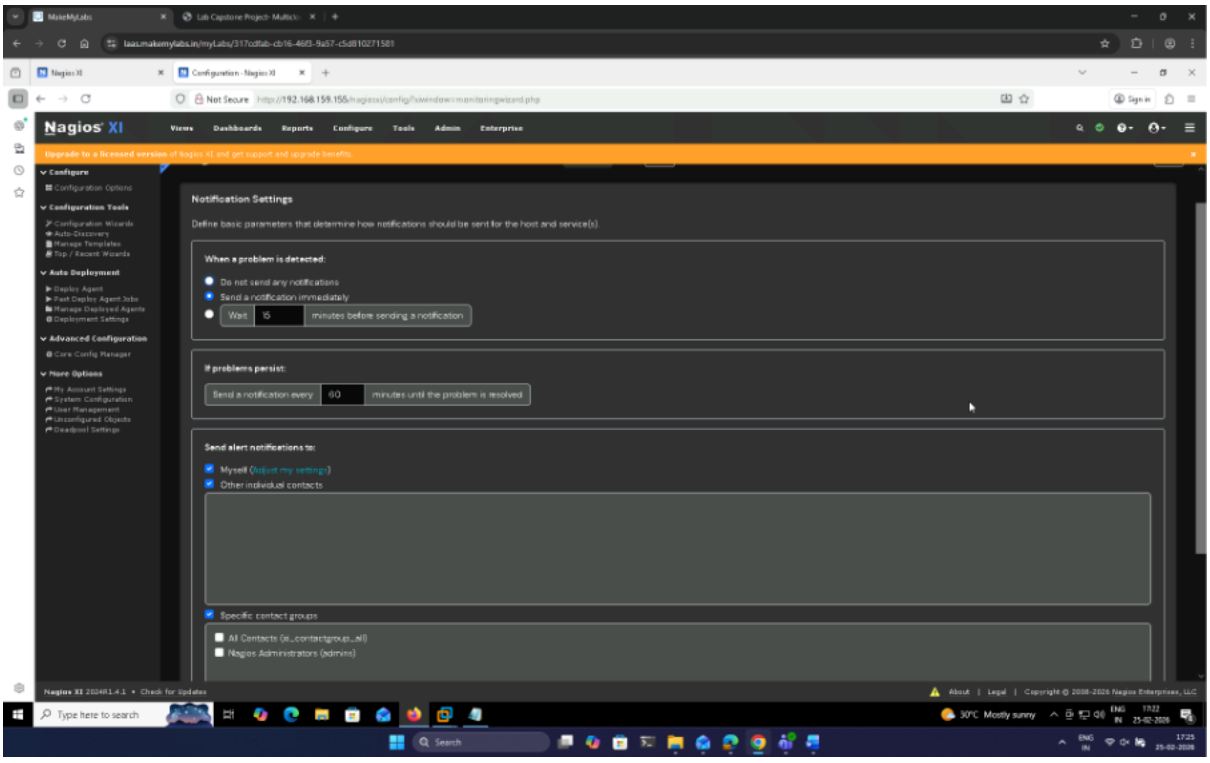
We have given: -

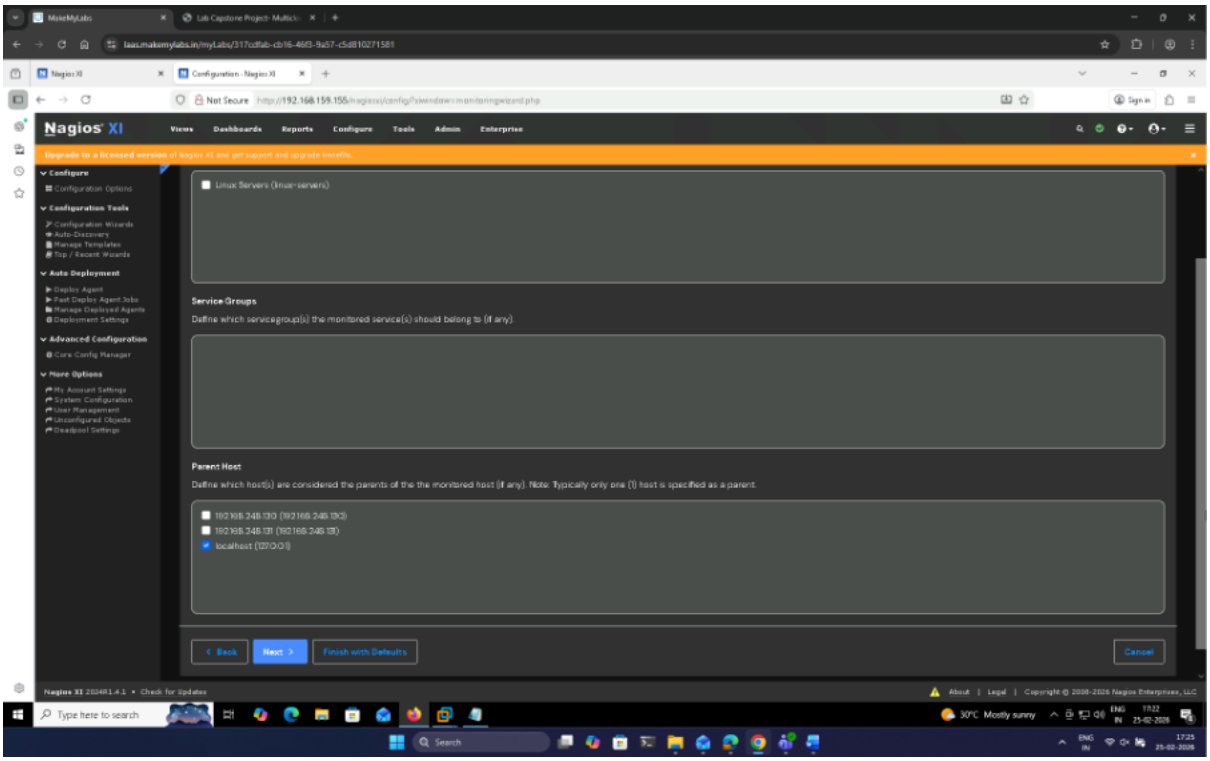
- CPU Monitoring: -
    - Warning: - 70%

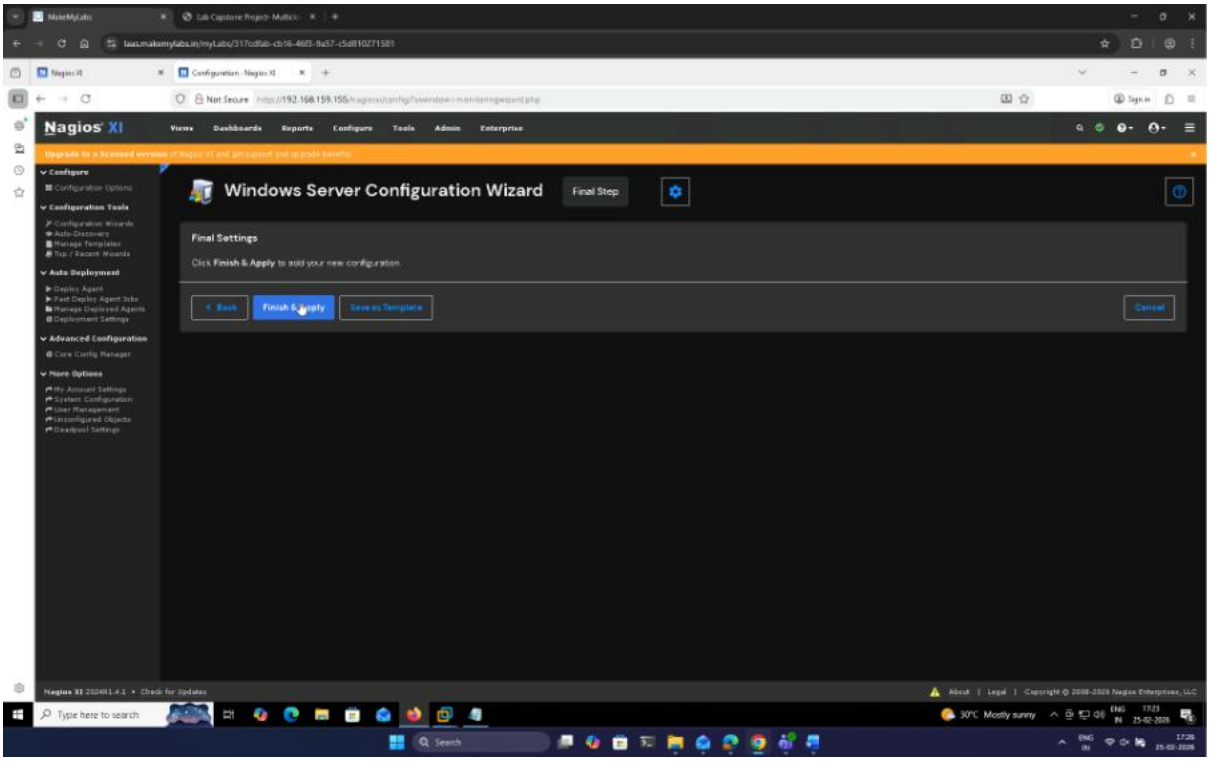
    - Critical: - 90%
- Memory Monitoring: -
    - Memory Usage: -
        - Warning: - 75%
        - Critical: - 90%
    - Swap Usage: -
        - Warning: - 60%
        - Critical: - 80%
- Disk Monitoring: -
    - Warning: - 85%
    - Critical: - 95%
- Network Interface Monitoring: -
    - Warning: - 50 MB/s
    - Critical: - 100 MB/s
- Windows Service Monitoring: -
    - Appinfo; EventLog; LanmanServer; W32Time; WinDefend – Running
- Process Monitoring: -
    - MsMpEng.exe (min – 1; max – 1); explorer.exe (min – 1; max – 1); svchost.exe (min – 5; max – 3)

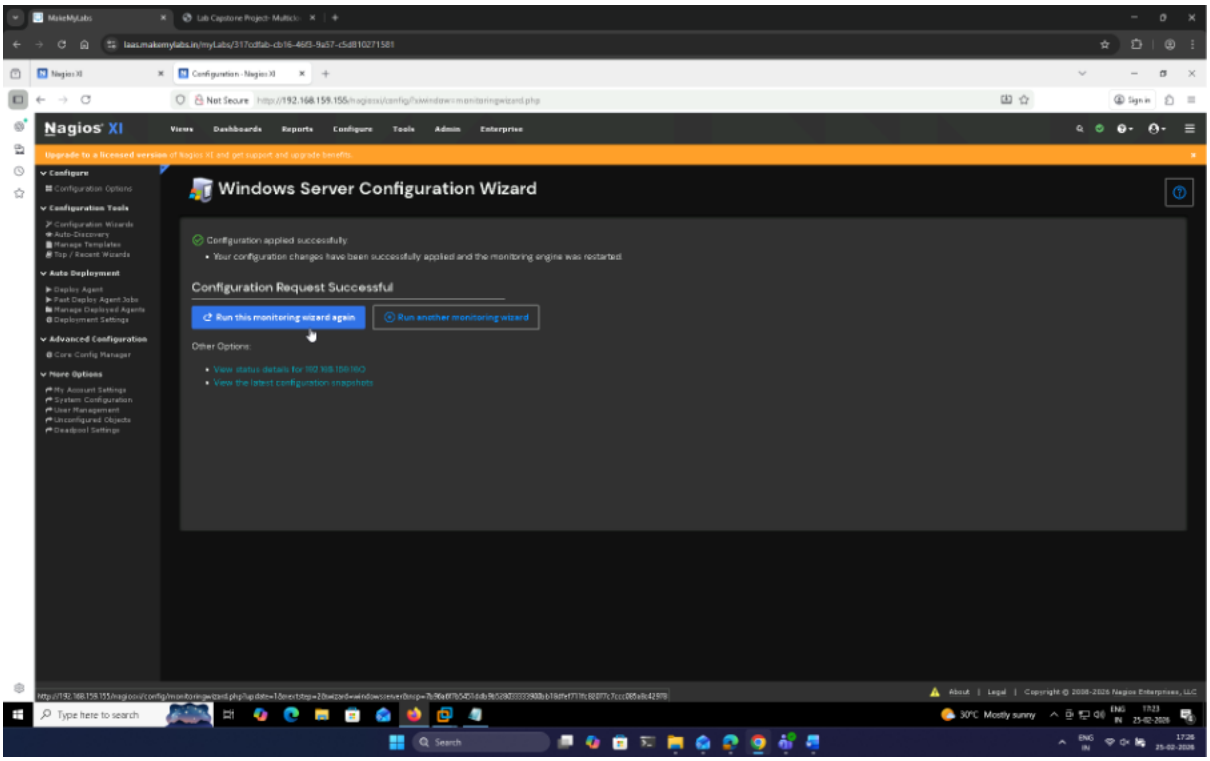
During configuration validation, the Windows Time (W32Time) service was corrected to an expected “Running” state to ensure accurate service health monitoring. Additionally, process monitoring thresholds were refined. Critical security and system processes such as MsMpEng.exe and explorer.exe were configured to trigger alerts if fewer than one instance is detected. Since svchost.exe typically runs multiple instances, appropriate minimum thresholds were defined to detect abnormal process termination without generating false positives. These refinements improve monitoring accuracy and align with real-world system behaviour.

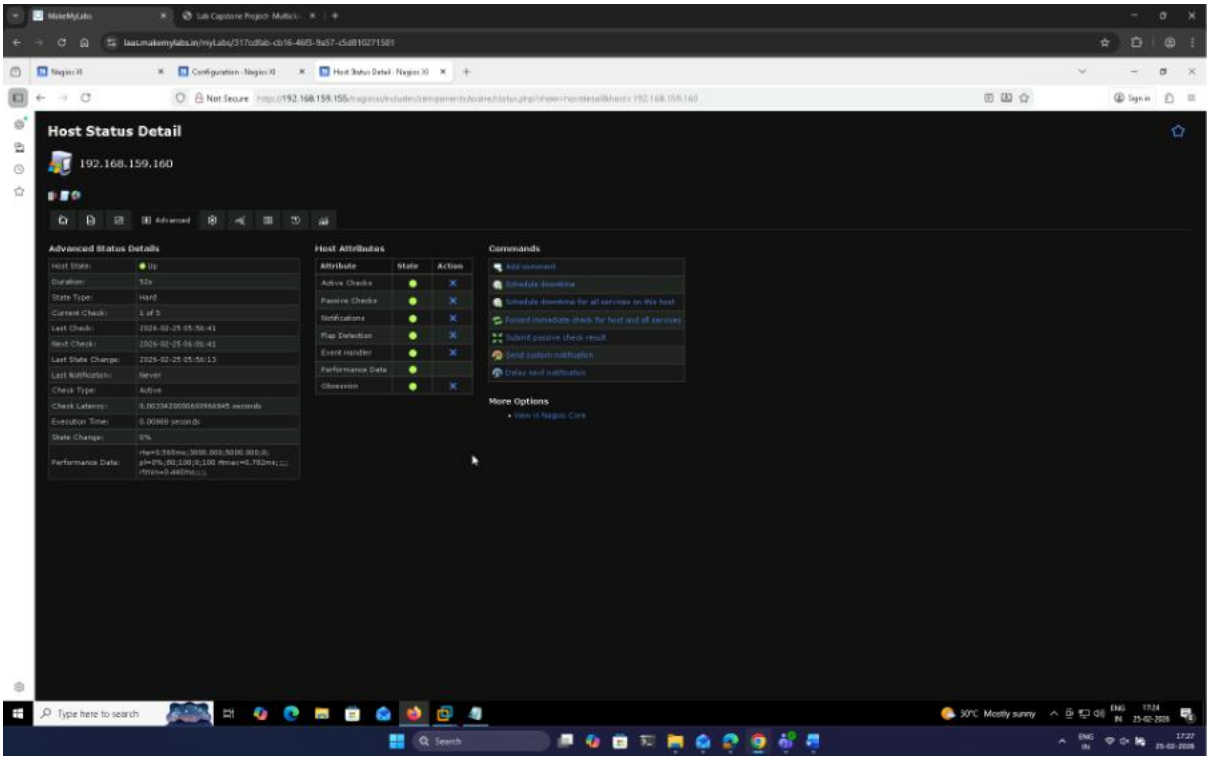

Nagios XI
Windows Server Configuration Wizard
Step 3
Monitoring Settings
Define basic parameters that determine how the host and service(s) should be monitored.
Under normal circumstances:
Monitor the host and service(s) every 5 minutes
When a potential problem is first detected:
Re-check the host and service(s) every 1 minutes up to 5 times before sending a notification
Back
Next >
Finish with Defaults
Cancel

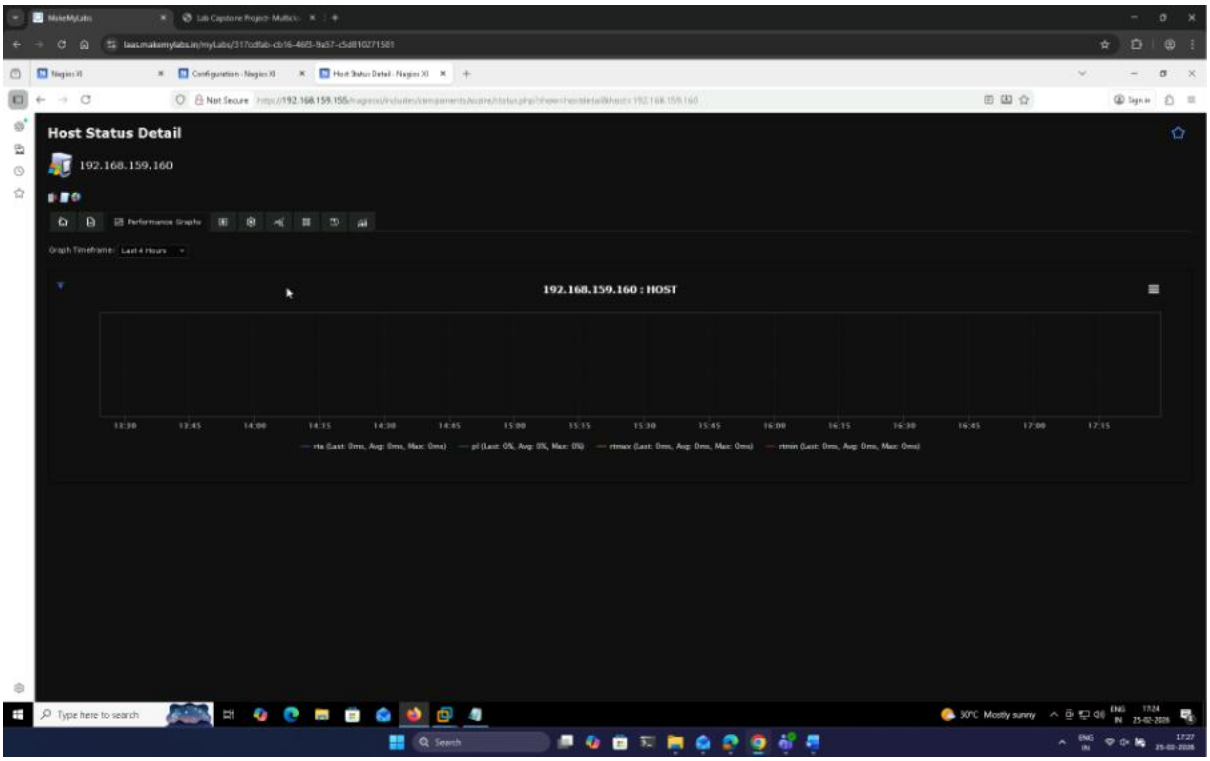

Nagios XI
Notification Settings
Define basic parameters that determine how notifications should be sent for the host and service(s).
When a problem is detected:
Do not send any notifications
Send a notification immediately
Wait 15 minutes before sending a notification
If problems persist:
Send a notification every 60 minutes until the problem is resolved
Send alert notifications to:
Myself (Adjust my settings)
Other individual contacts
Specific contact groups
All Contacts (ie_contactgroup_all)
Nagios Administrators (admins)

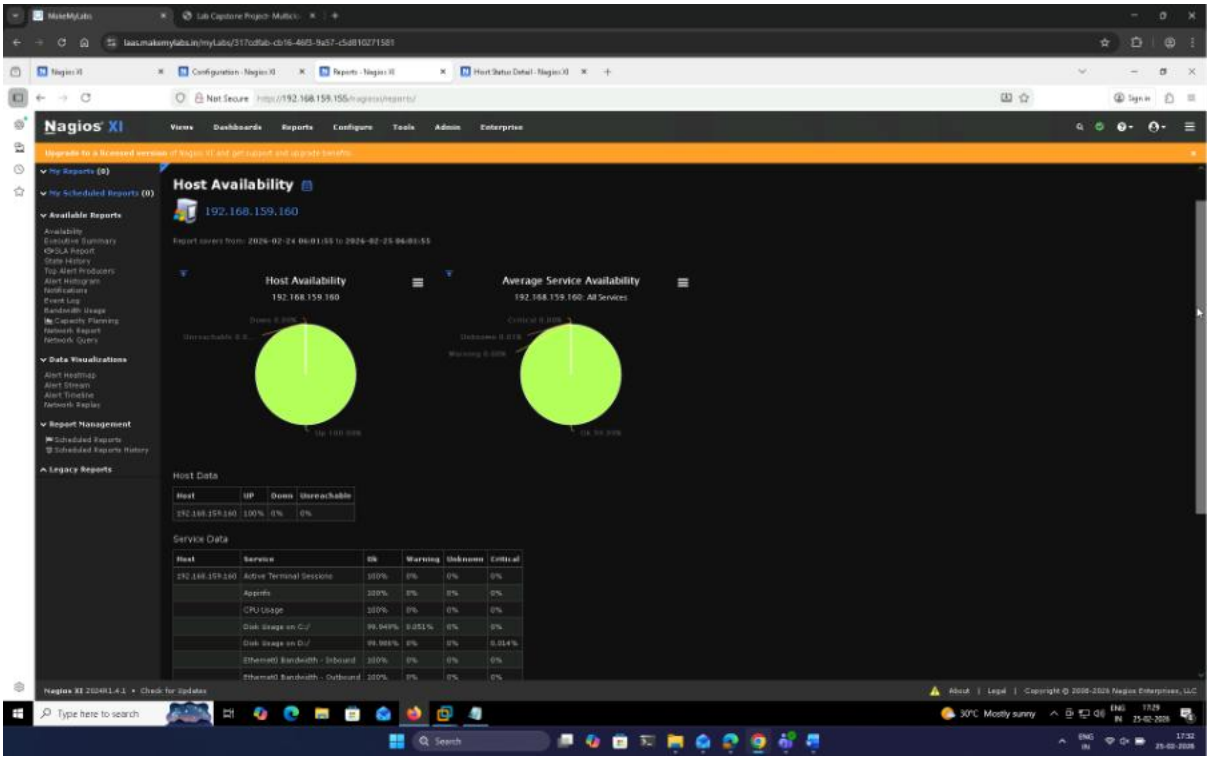

Nagios XI
Linux Servers (linux-servers)
Service Groups
Define which servicegroup(s) the monitored service(s) should belong to (if any).
Parent Host
Define which host(s) are considered the parents of the the monitored host (if any). Note: Typically only one (1) host is specified as a parent.
localhost (127.0.0.1)
< Back
Next >
Finish with Defaults
Cancel

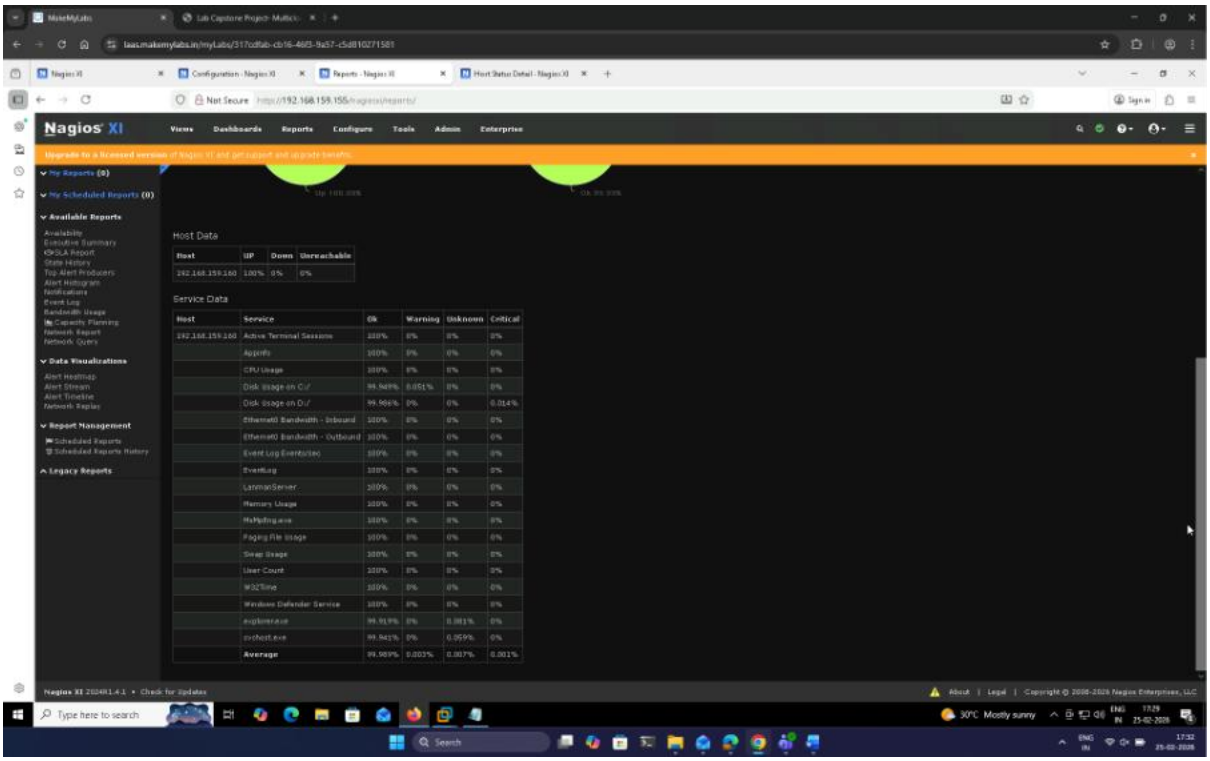

Nagios XI
Windows Server Configuration Wizard
Final Step
Final Settings
Click Finish & Apply to add your new configuration.
< Back
Finish & Apply
Save as Template
Cancel


Nagios XI
Windows Server Configuration Wizard
Configuration applied successfully
Your configuration changes have been successfully applied and the monitoring engine was restarted
Configuration Request Successful
Run this monitoring wizard again
Run another monitoring wizard
Other Options:


Host Status Detail
192.168.159.160
Advanced Status Details
Host Attributes
Commands
More Options


Host Status Detail
192.168.159.160
192.168.159.160 : HOST


Nagios XI
Host Availability
192.168.159.160
Host Availability
192.168.159.160
Average Service Availability
192.168.159.160: All Services
Host Data
Service Data



The Nagios XI monitoring configuration for APP-SRV-1 was successfully completed using the NCPA agent. Resource utilization thresholds, critical service monitoring, and process validation checks were implemented to ensure proactive system health monitoring.

Conclusion: -

In this Monitoring assignment, two enterprise-grade monitoring solutions – SolarWinds and Nagios XI were successfully implemented and configured to monitor network and system performance. Using SolarWinds, nodes were added to the monitoring console, credentials were configured, and application as well as system monitoring components were enabled. Alerts and notifications were validated to ensure proper detection of system health and service status.

Similarly, Nagios XI was deployed with the NCPA agent to monitor a Windows 10 Client machine. Resource thresholds were defined for CPU, Memory, Disk Usage, Network Interfaces, Critical Services, and essential processes. Secure communication between the Nagios server and client was established, and alert generation was verified through threshold testing.

Through both implementations, the lab demonstrated practical experience in centralized monitoring, agent-based monitoring, performance threshold configuration, service validation, and proactive alerting. Overall, the exercise reinforced the importance of

continuous infrastructure monitoring to ensure system reliability, availability, and early detection of potential failures in enterprise environments.